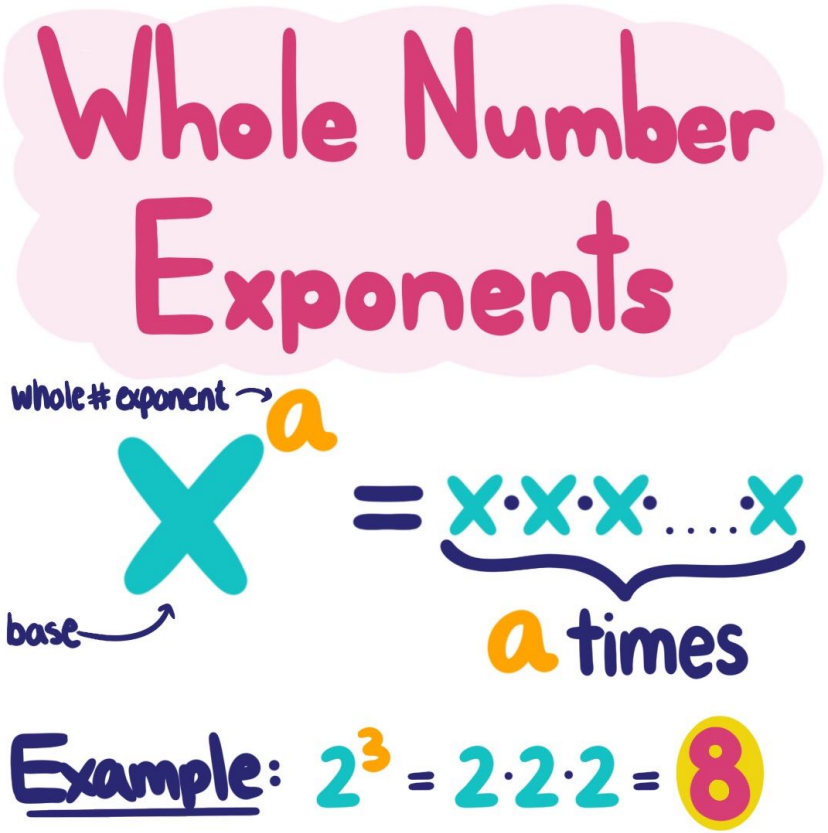
Quantum Information Systems

Brian Rashap

July 2026

Classical Bits

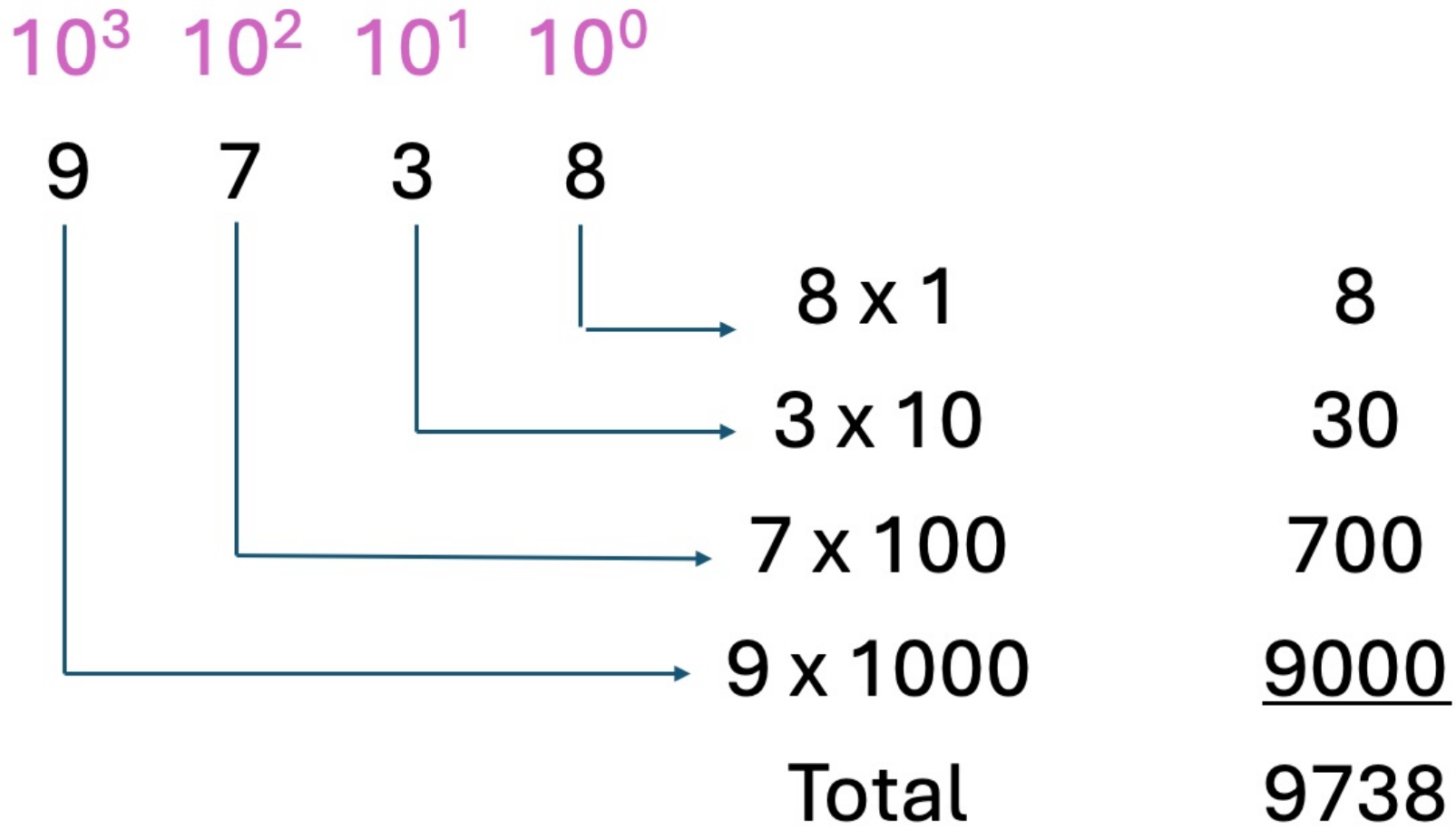
Exponents



The rule for zero as an exponent:

Any nonzero real number raised to the power of zero is one, this means anything that looks like x^0 will always equal 1 if x is not equal to zero.

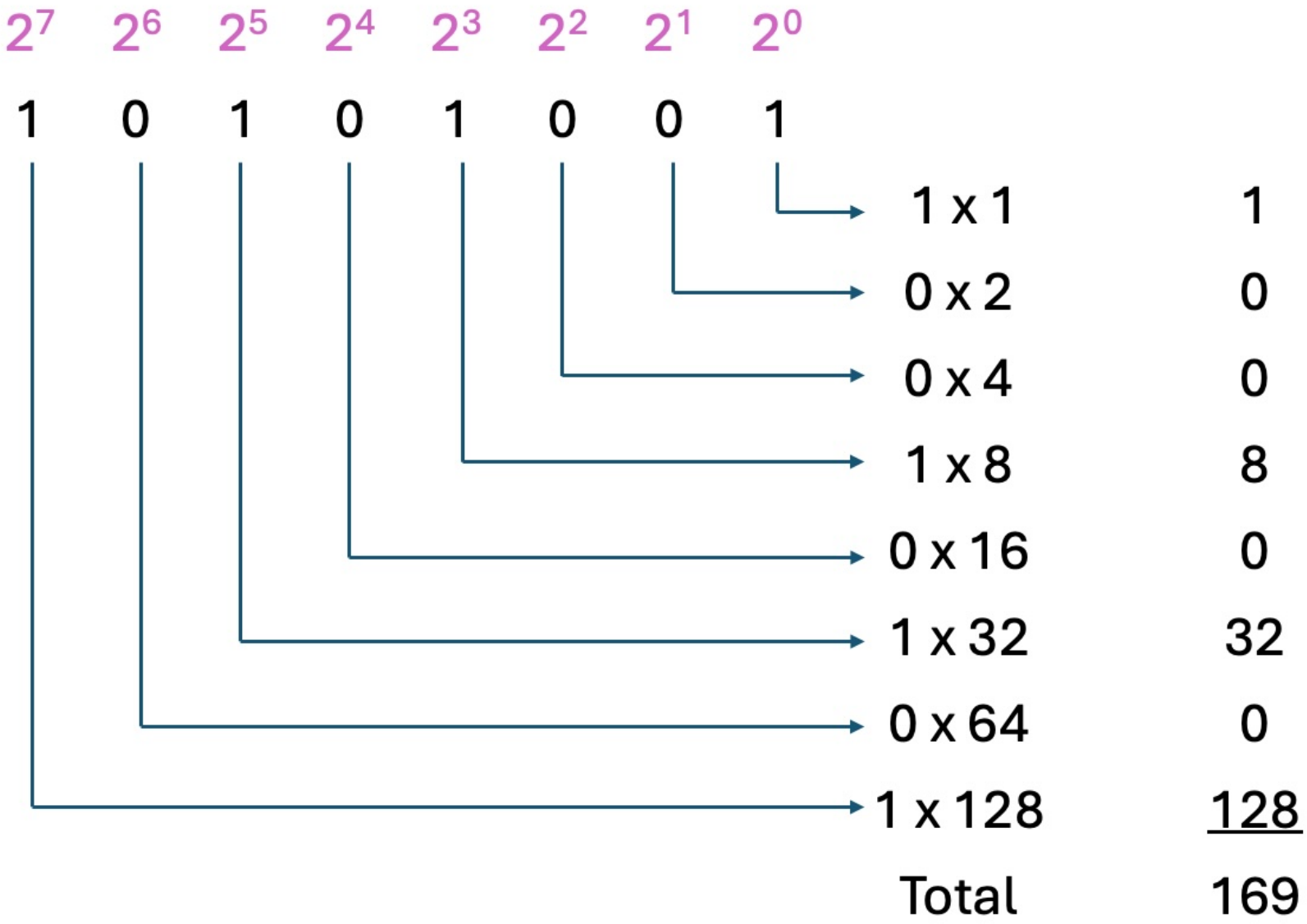
Decimal


10^3 10^2 10^1 10^0
9 7 3 8
8 x 1
3 x 10
7 x 100
9 x 1000
8
30
700
9000
Total
9738

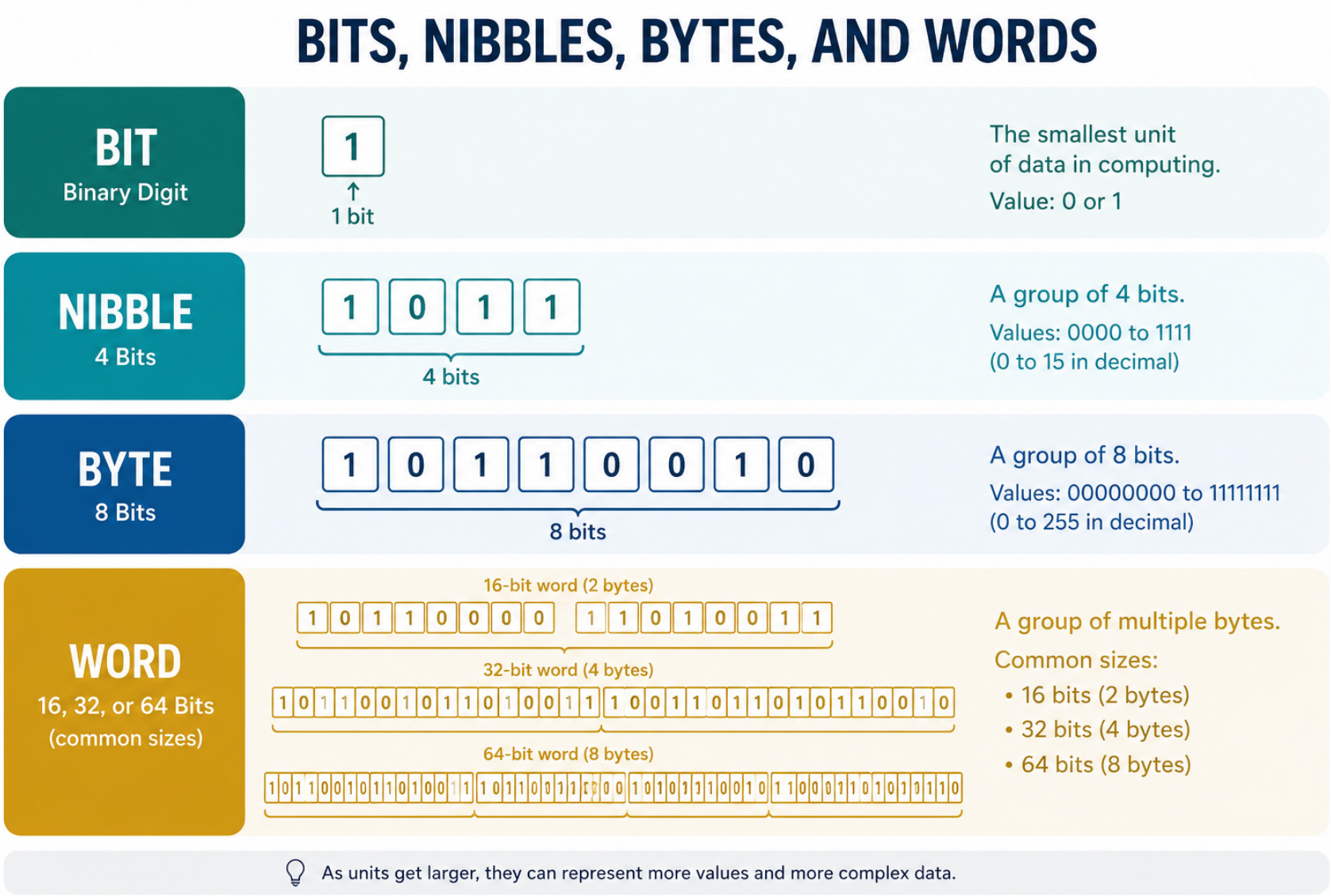
A Bit by any other name

	OFF	False	0
	ON	True	1

Binary


2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0
1 0 1 0 1 0 0 1
1 x 1 1
0 x 2 0
0 x 4 0
1 x 8 8
0 x 16 0
1 x 32 32
0 x 64 0
1 x 128 128
Total 169

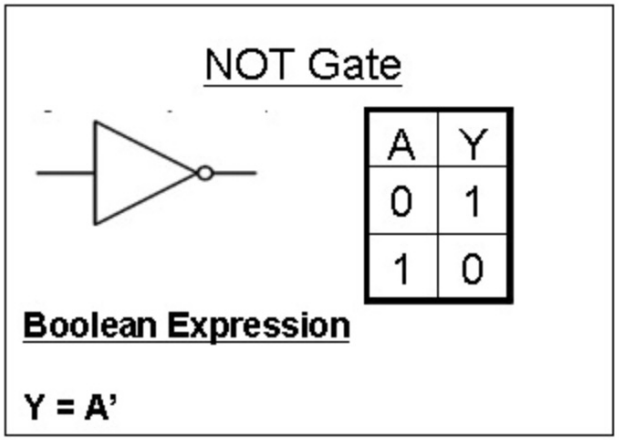
Bits, Nibbles, Bytes, Words



Classical Gates

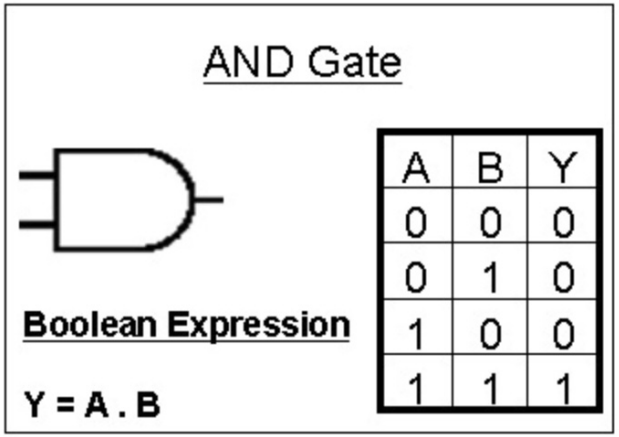
NOT Gate

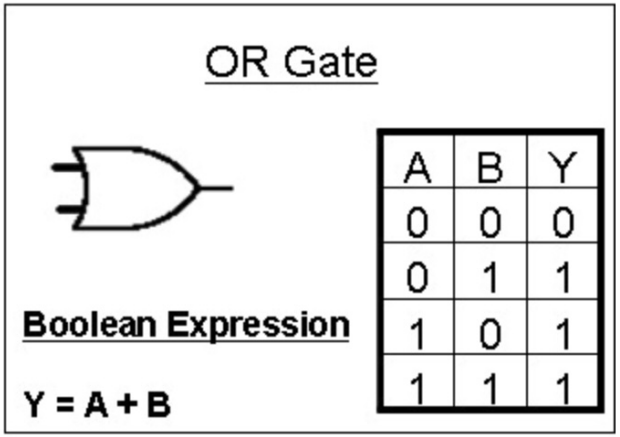
- The NOT gate take the input bit and flips it
  - 0 becomes a 1
  - 1 becomes a 0



AND and OR Gate

- The AND gate returns a 1 if both inputs are 1's.
- The OR gate returns a 1 if either input is a 1.





NAND and NOR Gate

The NOT AND (NAND) and the NOT OR (NOR) gates behave like the AND or OR gate followed by a NOT.

- The NAND gate returns a 1 if either input is a 0.
- The NOR gate returns a 1 if both inputs are a 0.

NAND Gate

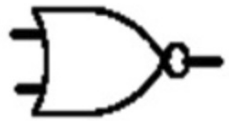


A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Boolean Expression

Y = (A . B)' = A' + B'

NOR Gate



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Boolean Expression

Y = (A + B)'

XOR and XNOR Gate

The exclusive (X) OR and NOR gates, are similar to the OR and NOR but only if either/or but not both inputs are a 1 or 0 respectively.

XOR Gate



Boolean Expression

$Y = A \oplus B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

XNOR Gate



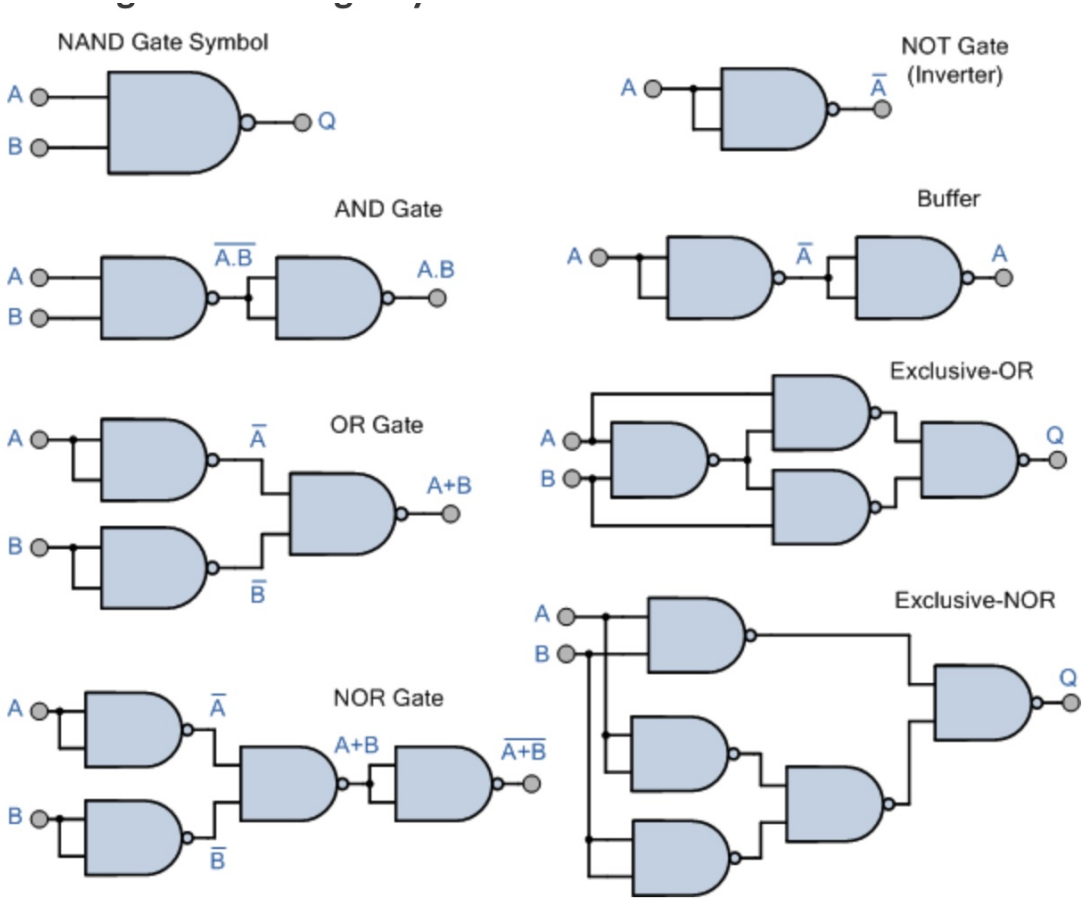
Boolean Expression

$Y = A \odot B$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Universal Gates

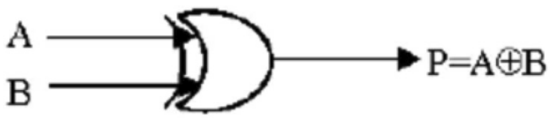
ALL other logic gate functions can be created using only NAND gates making it a universal logic gate.



Reversible Gates

The classical gates (other than NOT) are not reversible gates (e.g., the inputs can not be inferred from the output). However, it is possible to make them reversible by adding an additional output.

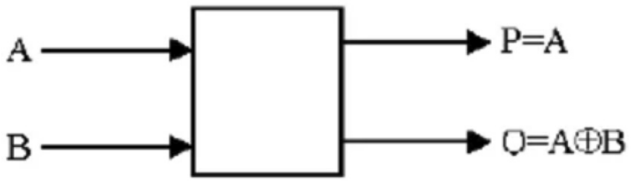
A	B	P
0	0	0
0	1	1
1	0	1
1	1	0



Irreversible EX-OR Gate

Garbage Output

A	B	P	Q
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0



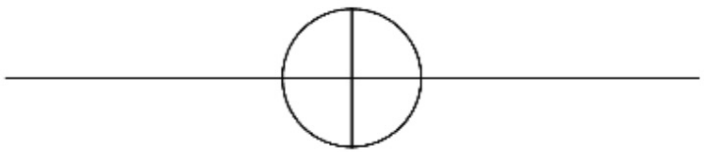
Reversible EX-OR Gate

Quantum Gates - A simplified Introduction

The Quantum NOT Gate

We are going to move from bits with values 0 or 1, to quantum bits (or qubits) represented as $|\psi\rangle$ with states $|0\rangle$ and $|1\rangle$.

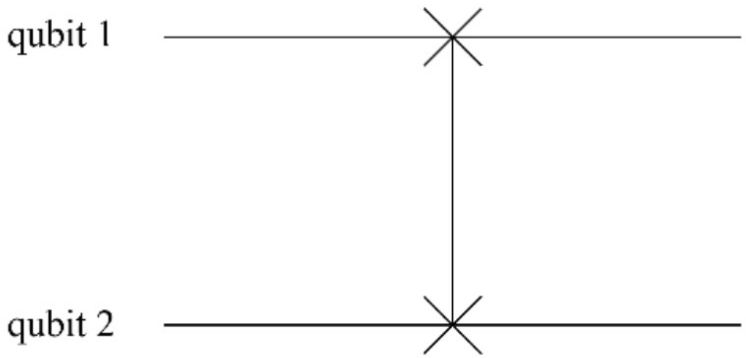
The quantum NOT gate is the analogue of the classical NOT gate used in quantum computing. It turns a state of $|0\rangle$ into a state of $|1\rangle$, and vice versa.



Note, it is required that all quantum gates be reversible.

The Quantum SWAP Gate

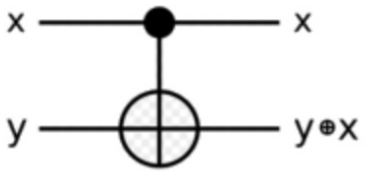
The SWAP gate is the first of the two input quantum gates.

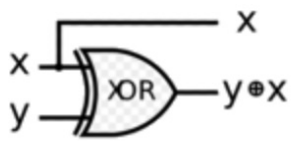


In this case $|\psi_1\rangle$ swaps values with $|\psi_2\rangle$.

The Quantum CNOT Gate

For Controlled NOT (or CNOT) gate, consider qubits x and y, where x is the control qubit and y is the target qubit. The CNOT gate applies a NOT to the target qubit only if the control qubit is $|1\rangle$, otherwise the target is unchanged. Additionally, the control qubit is passed through always unchanged. This is the quantum equivalent of the classical XOR gate made reversible.



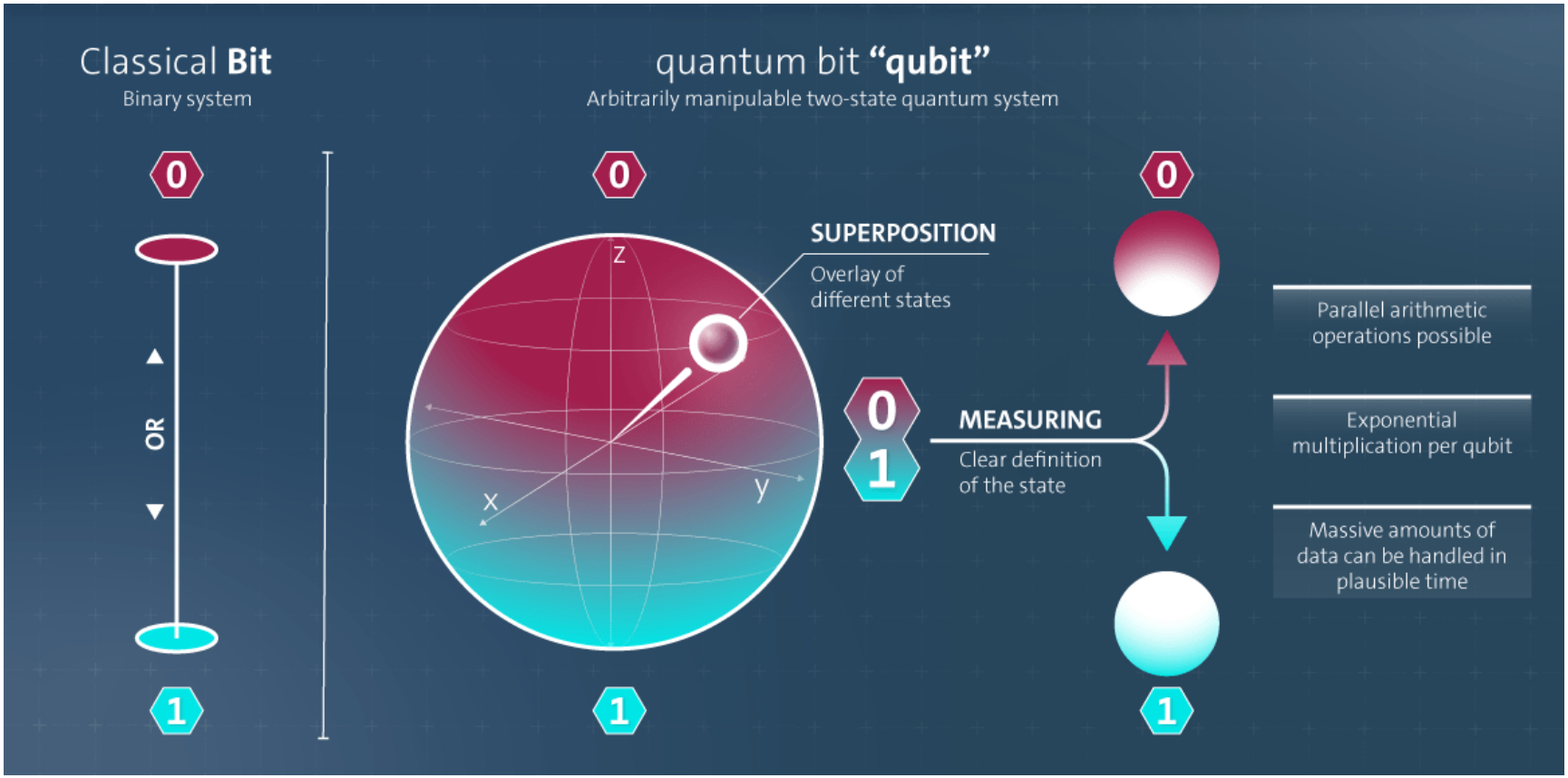


input		output	
x	y	x	y+x
$\|0\rangle$	$\|0\rangle$	$\|0\rangle$	$\|0\rangle$
$\|0\rangle$	$\|1\rangle$	$\|0\rangle$	$\|1\rangle$
$\|1\rangle$	$\|0\rangle$	$\|1\rangle$	$\|1\rangle$
$\|1\rangle$	$\|1\rangle$	$\|1\rangle$	$\|0\rangle$

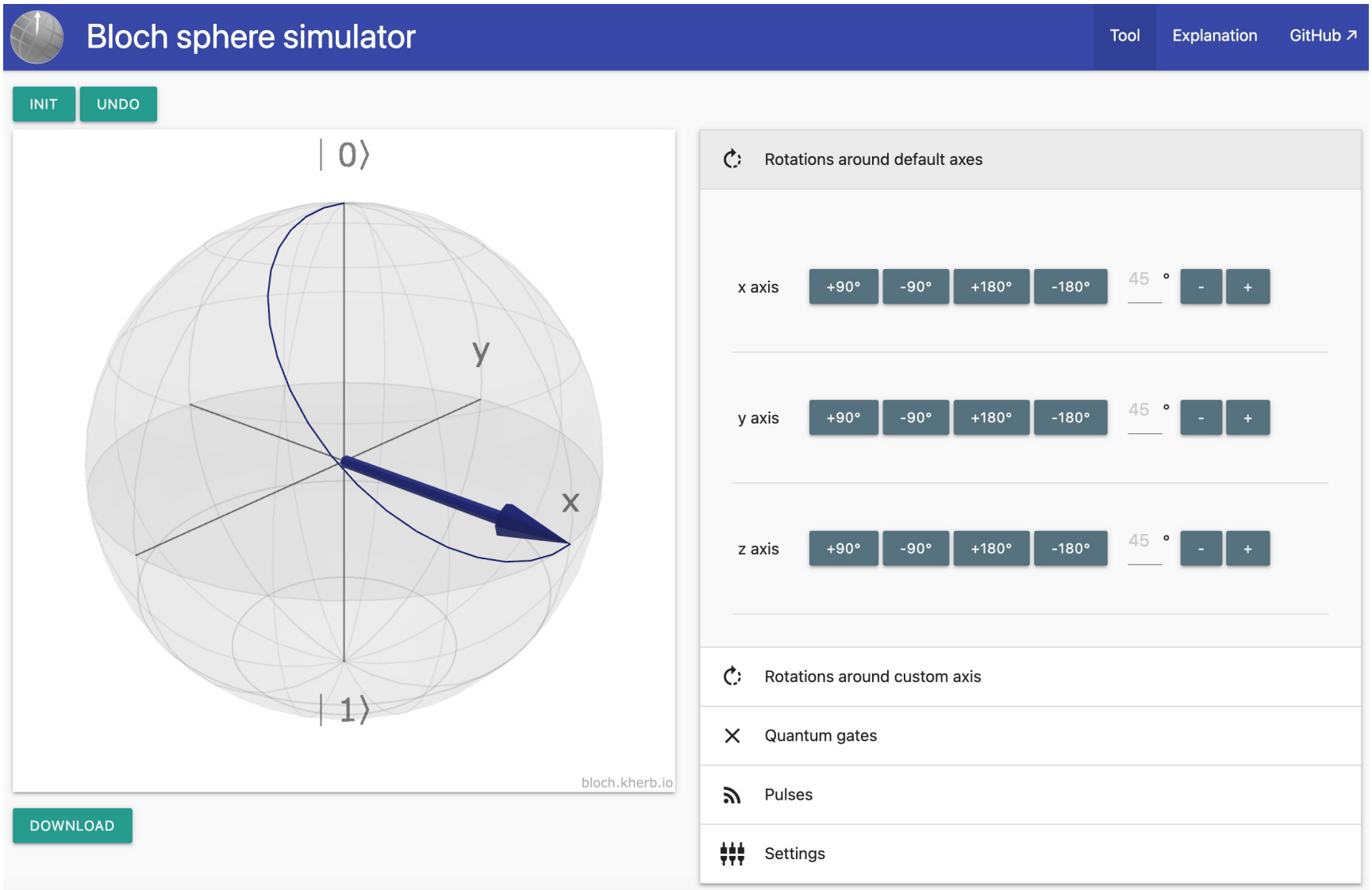
input		output	
x	y	x	y+x
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

What is a Qubit

Bit vs Qubit


Classical Bit
Binary system
0
OR
1
quantum bit "qubit"
Arbitrarily manipulable two-state quantum system
0
z
x
y
1
SUPERPOSITION
Overlay of different states
0
1
MEASURING
Clear definition of the state
0
1
Parallel arithmetic operations possible
Exponential multiplication per qubit
Massive amounts of data can be handled in plausible time

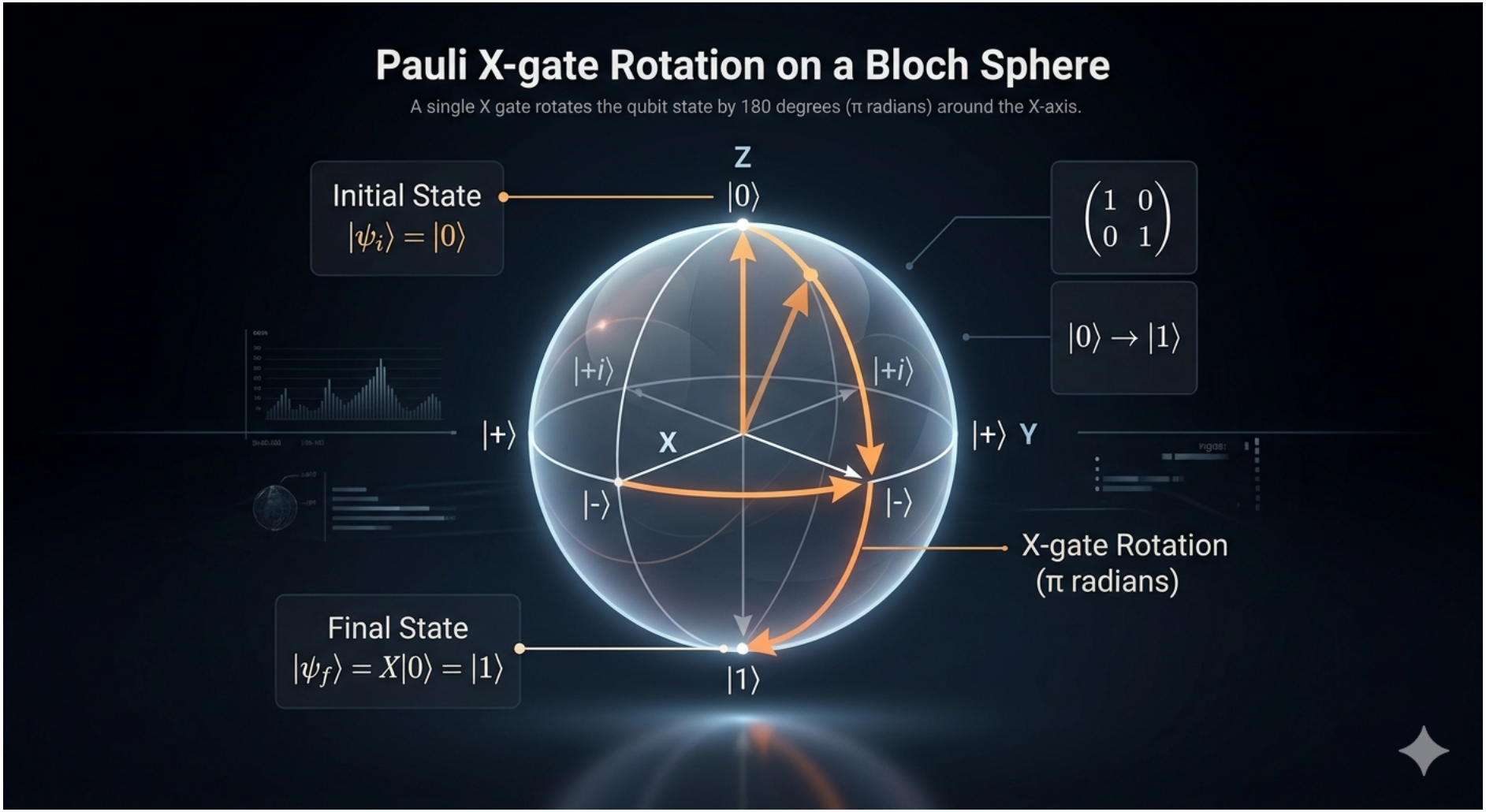
Bloch Sphere



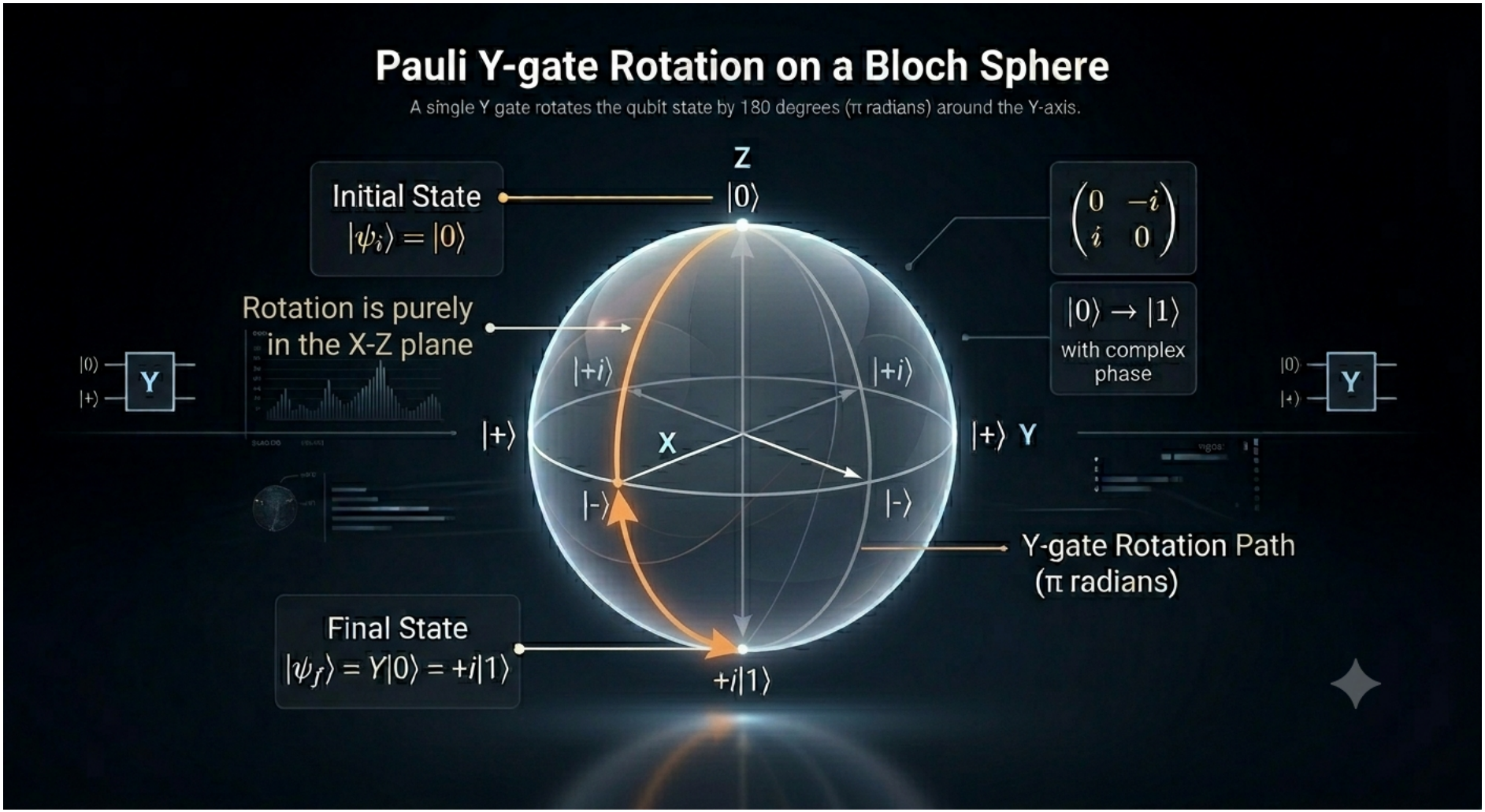
https://bloch.kherb.io/

Quantum Gates

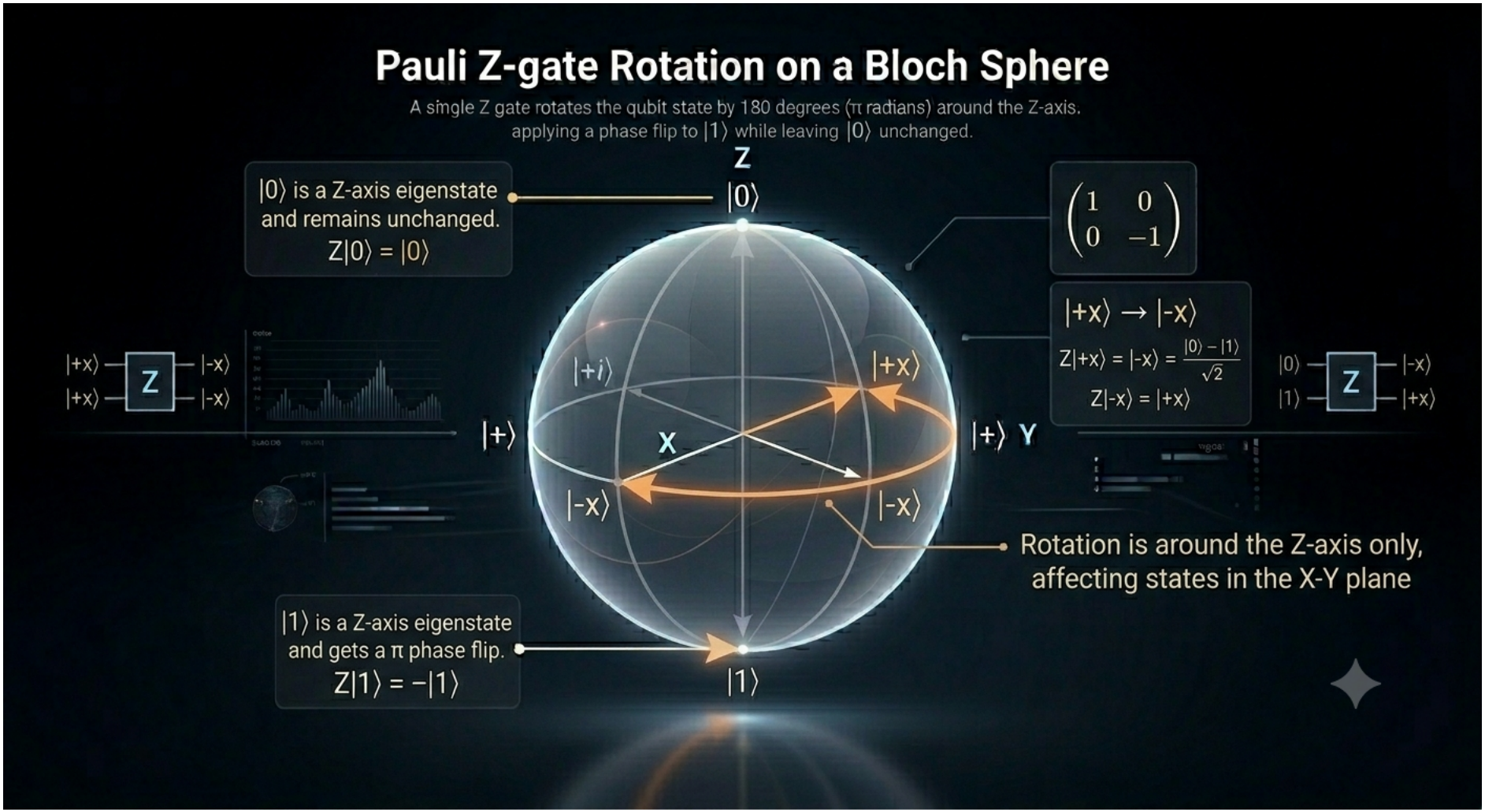
Pauli X Gate


Pauli X-gate Rotation on a Bloch Sphere
A single X gate rotates the qubit state by 180 degrees (π radians) around the X-axis.
Initial State
$|\psi_i\rangle = |0\rangle$
Z
$|0\rangle$
$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
$|0\rangle \rightarrow |1\rangle$
$|+i\rangle$
$|+i\rangle$
$|+\rangle$
X
$|+\rangle$ Y
$|-\rangle$
$|-\rangle$
X-gate Rotation
(π radians)
Final State
$|\psi_f\rangle = X|0\rangle = |1\rangle$
$|1\rangle$

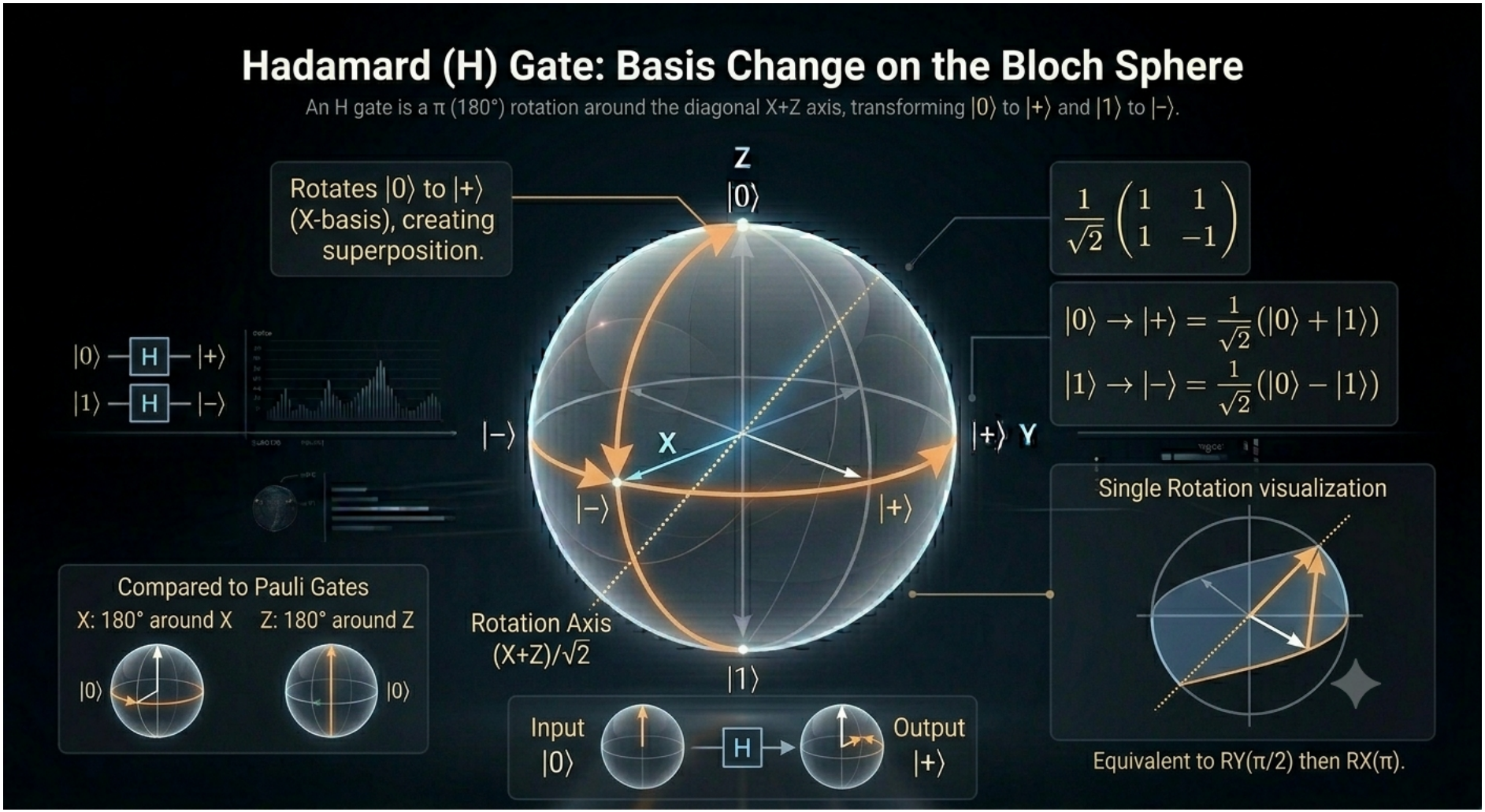
Pauli Y Gate


Pauli Y-gate Rotation on a Bloch Sphere
A single Y gate rotates the qubit state by 180 degrees (π radians) around the Y-axis.
Initial State
$|\psi_i\rangle = |0\rangle$
Z
$|0\rangle$
$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$
Rotation is purely in the X-Z plane
$|0\rangle \rightarrow |1\rangle$
with complex phase
$|0\rangle$
$|+\rangle$
Y
$|+i\rangle$
$|+i\rangle$
$|+\rangle$
X
$|+\rangle$ Y
$|-\rangle$
$|-\rangle$
Y-gate Rotation Path
(π radians)
Final State
$|\psi_f\rangle = Y|0\rangle = +i|1\rangle$
$+i|1\rangle$

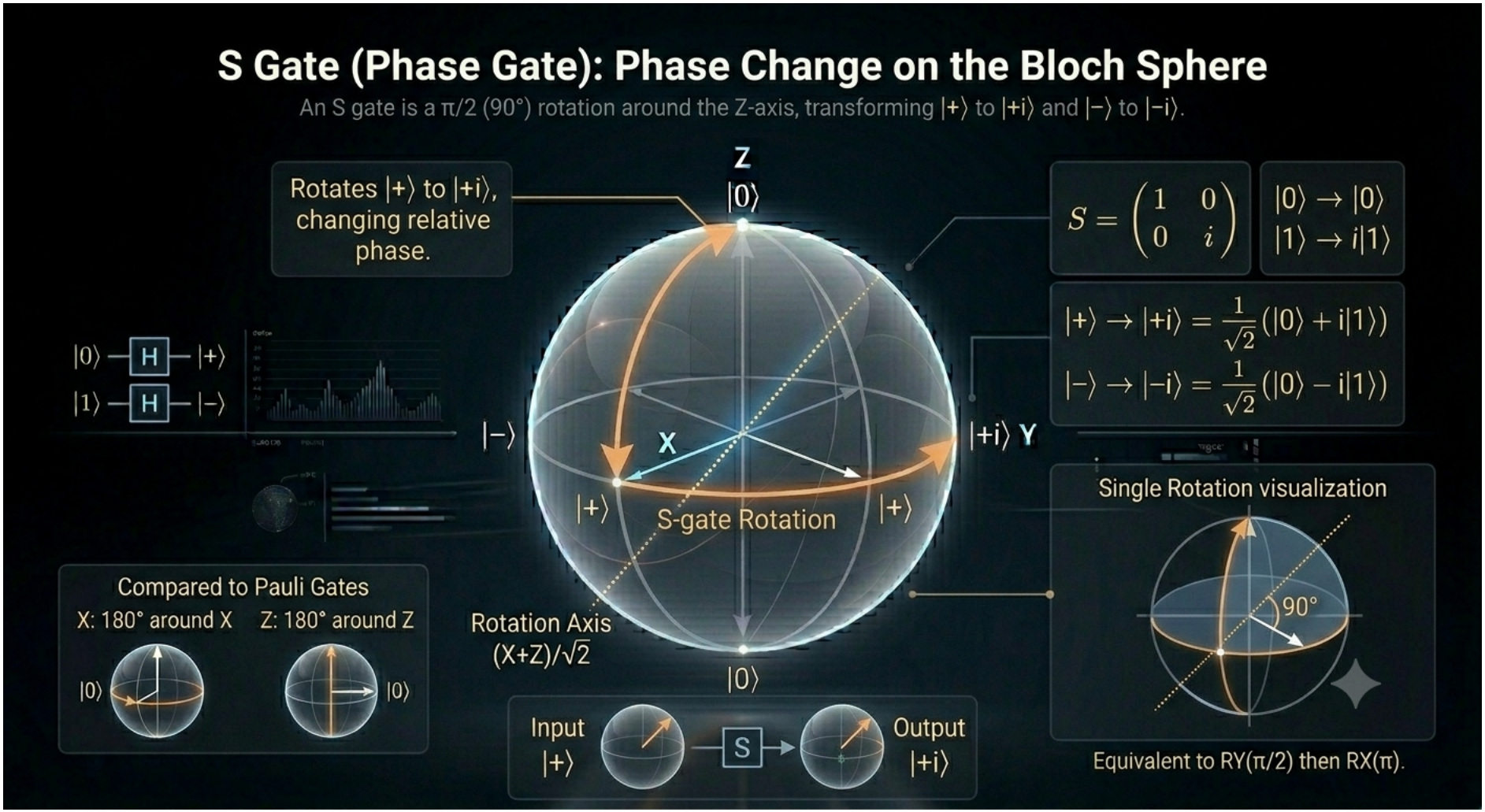
Pauli Z Gate


Pauli Z-gate Rotation on a Bloch Sphere
A single Z gate rotates the qubit state by 180 degrees (π radians) around the Z-axis, applying a phase flip to |1⟩ while leaving |0⟩ unchanged.
|0⟩ is a Z-axis eigenstate and remains unchanged.
Z|0⟩ = |0⟩
Z
|0⟩
$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
|+x⟩ → |-x⟩
Z|+x⟩ = |-x⟩ = $\frac{|0⟩ - |1⟩}{\sqrt{2}}$
Z|-x⟩ = |+x⟩
|+x⟩ Z |-x⟩
|+x⟩ |-x⟩
|0⟩ Z |-x⟩
|1⟩ |+x⟩
|+i⟩
|+x⟩
|+⟩
X
|+⟩ Y
|-x⟩
|-x⟩
Rotation is around the Z-axis only, affecting states in the X-Y plane
|1⟩ is a Z-axis eigenstate and gets a π phase flip.
Z|1⟩ = −|1⟩
|1⟩

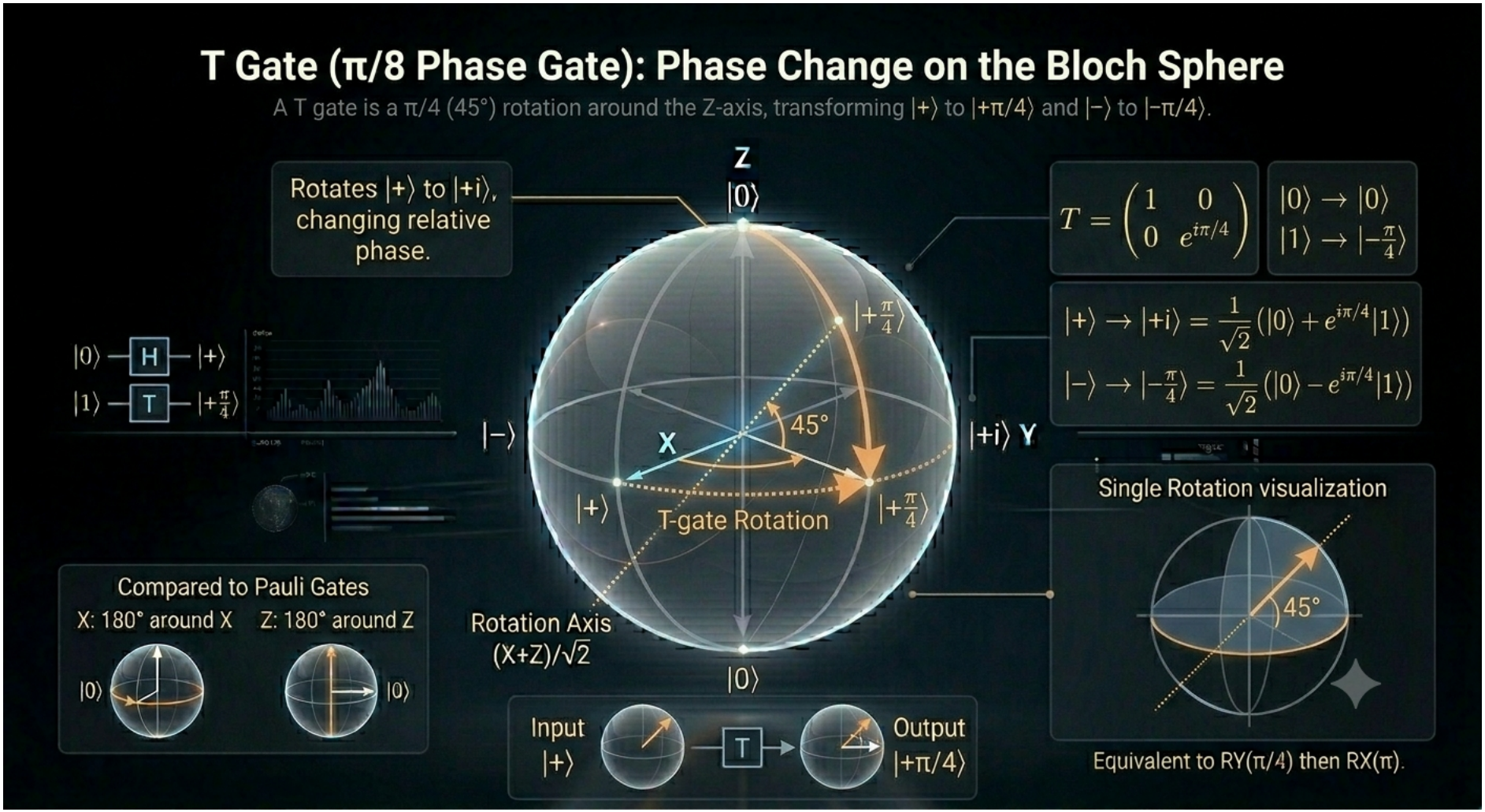
Hadamard Gate



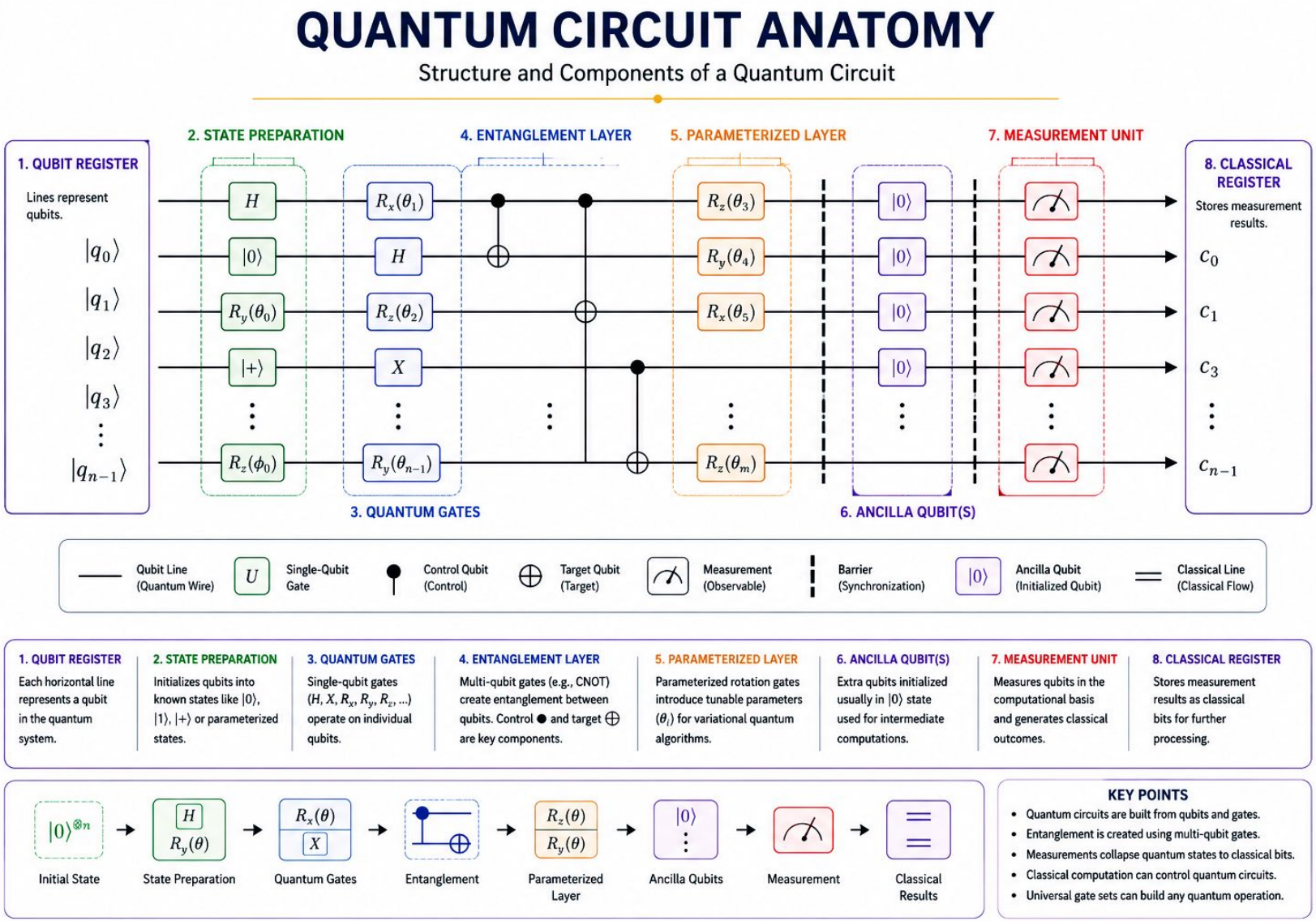
S Gate or Phase Gate



T Gate or $\frac{\pi}{8}$ Phase Gate



Anatomy of a Quantum Circuit



Linear Algebra

Quantum States - Column Vectors

Using Dirac's bra-ket notation, we write the quantum states using the ket to represent a column vector:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

The quantum state $|\psi\rangle$ is the superposition of these two states with amplitudes α and β:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle = \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

Quantum States - Row Vectors

The transpose of a column vector is obtained by writing it as a row vector. But, in quantum computing, we use the conjugate transpose:

$$\begin{bmatrix}\alpha \\ \beta\end{bmatrix}^\dagger = \begin{bmatrix}\alpha^* \beta^*\end{bmatrix}$$

This can be written using the bra:

$$|\psi\rangle^\dagger = \langle\psi| = \begin{bmatrix}\alpha^* \beta^*\end{bmatrix}$$

Going between kets and bras:

$$|\psi\rangle = \begin{bmatrix}\alpha \\ \beta\end{bmatrix} \iff \langle\psi| \begin{bmatrix}\alpha^* \beta^*\end{bmatrix}$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \iff \langle\psi| = \alpha^* \langle 0| + \beta^* \langle 1|$$



Inner Product

Consider two states:

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, |\phi\rangle = \begin{bmatrix} \gamma \\ \delta \end{bmatrix}$$

One way to multiple them is the inner product:

$$\langle\psi|\,|\phi\rangle = \langle\psi|\,\phi\rangle = \begin{bmatrix} \alpha^* \beta^* \end{bmatrix} \begin{bmatrix} \gamma \\ \delta \end{bmatrix} = \alpha^*\gamma + \beta^*\delta$$

Note that is inner product is just a scalar.

Orthonormality

- Take the inner product of $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ with itself

$$\langle\psi|\,\psi\rangle = \begin{bmatrix}\alpha^* & \beta^*\end{bmatrix}\begin{bmatrix}\alpha \\ \beta\end{bmatrix} = |\alpha|^2 + |\beta|^2 = 1$$

  $\langle\psi|\,\psi\rangle$ is the total probability and if it is equal to 1 then the state $|\psi\rangle$ is considered normalized.

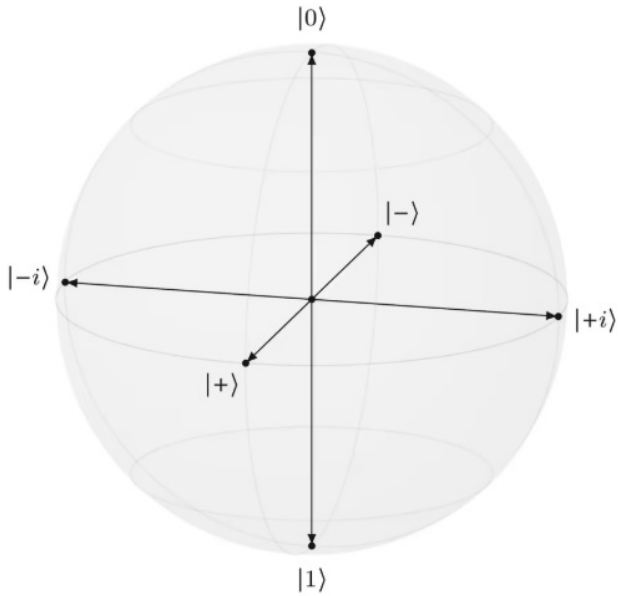
- Consider the inner product of the Z-basis states $|0\rangle$ and $|1\rangle$:

$$\langle 0|\,1\rangle = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}0 \\ 1\end{bmatrix} = 1 * 0 + 0 * 1 = 0$$

  These, and any two states with zero for the inner product, are considered orthogonal.

- As $|0\rangle$ and $|1\rangle$ are also normalized, they are considered orthonormal.

Orthonormality Continued



- Consider the inner product of the X-basis states $|+\rangle$ and $|-\rangle$:

$$\langle +|-\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}1\\-1\end{bmatrix} = \frac{1}{2}(1-1) = 0$$

- Consider the inner product of the Y-basis states $|i\rangle$ and $|-i\rangle$:

$$\langle i|-i\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & -i\end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix}1\\-i\end{bmatrix} = \frac{1}{2}\begin{bmatrix}1 & -i\end{bmatrix}\begin{bmatrix}1\\-i\end{bmatrix} = \frac{1}{2}(1-i^2) = 0$$

Matrices, Multiplication and Quantum Gates

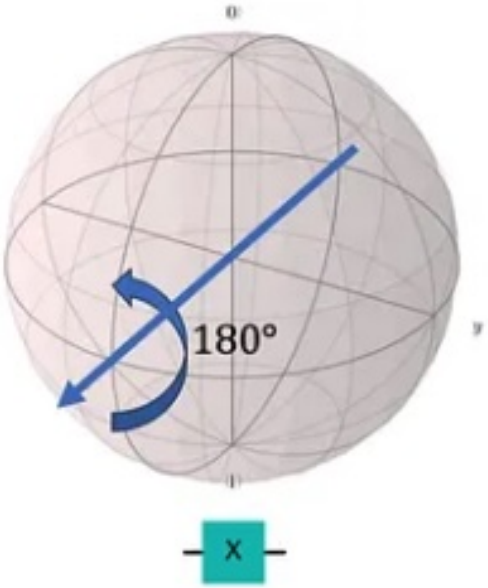
- Multiply a matrix and a vector:

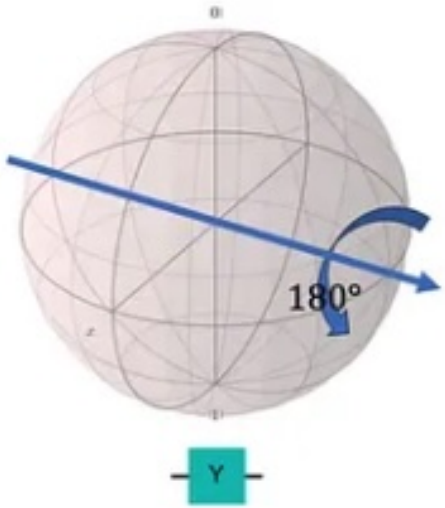
$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} a\alpha + b\beta \\ c\alpha + d\beta \end{bmatrix}$$

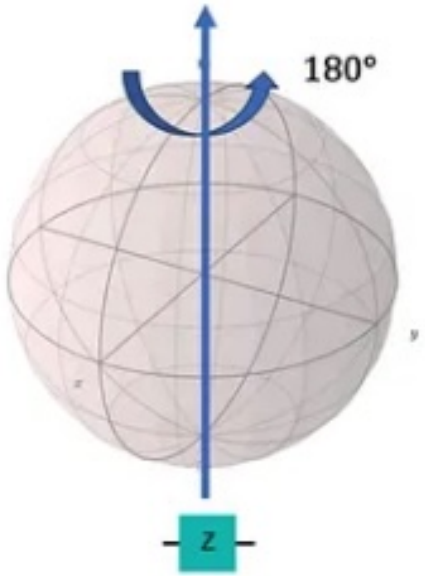
- Hadamard Gate applied to $|0\rangle$:

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$$

One Qubit Pauli Rotation Gates





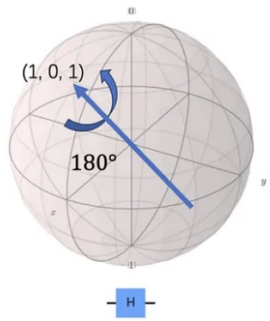


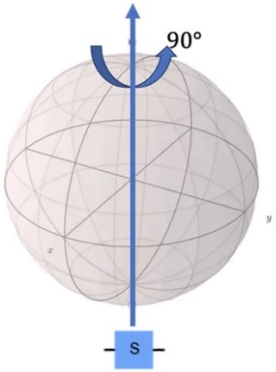
$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

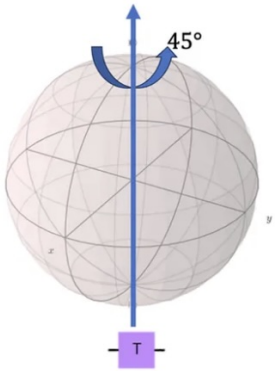
$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

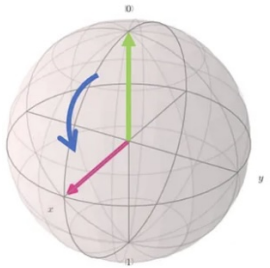
$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

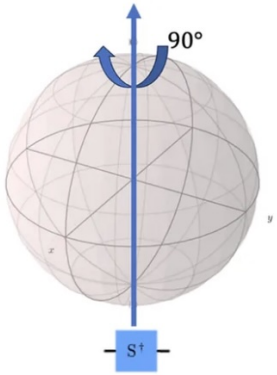
More One Qubit Gates

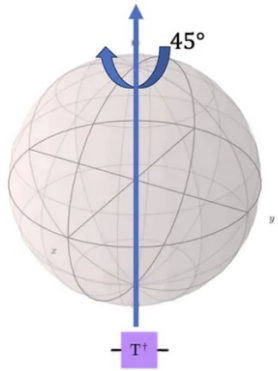












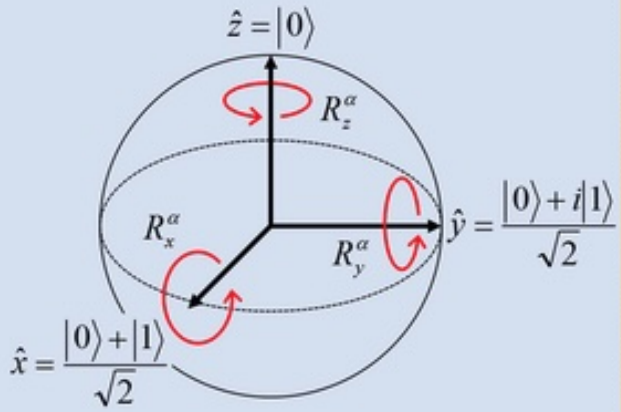
$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$S = \begin{bmatrix} 1 & 0 \\ 0 & e^{\frac{i\pi}{2}} \end{bmatrix}$$

$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{\frac{i\pi}{4}} \end{bmatrix}$$

Generic Gate Rotations

	input state	circuit symbol	output state	operator depictions and representations
Qubit rotations	$\|q_1\rangle$	R_z^α	$\|q_2\rangle = R_z^\alpha \|q_1\rangle$	$R_z^\alpha \equiv e^{-i\alpha Z/2} = \begin{pmatrix} e^{-i\alpha/2} & 0 \\ 0 & e^{+i\alpha/2} \end{pmatrix}$
	$\|q_1\rangle$	R_y^α	$\|q_2\rangle = R_y^\alpha \|q_1\rangle$	$R_y^\alpha \equiv e^{-i\alpha Y/2} = \begin{pmatrix} \cos(\alpha/2) & -\sin(\alpha/2) \\ \sin(\alpha/2) & \cos(\alpha/2) \end{pmatrix}$
	$\|q_1\rangle$	R_x^α	$\|q_2\rangle = R_x^\alpha \|q_1\rangle$	$R_x^\alpha \equiv e^{-i\alpha X/2} = \begin{pmatrix} \cos(\alpha/2) & -i\sin(\alpha/2) \\ -i\sin(\alpha/2) & \cos(\alpha/2) \end{pmatrix}$





Two Qubit Gates

The state of multiple qubits is written as the tensor product:

$$|0\rangle \oplus |0\rangle = |0\rangle |0\rangle = |00\rangle$$

For two qubits the z-basis is $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ with the superposition written as:

$$c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle$$

In the linear algebra representation, the tensor product is the Kronecker product:

$$|00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, |01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, |10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, |11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

The Controlled NOT (CNOT) Gate

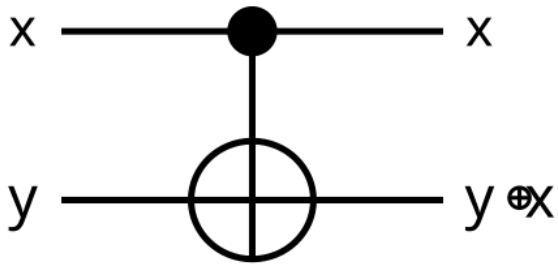
Given the two qubit system $|q_0 q_1\rangle$ and Pauli-X gate (the NOT operation) is applied to the q_1 qubit if $q_0 = |1\rangle$, otherwise q_1 is unchanged.

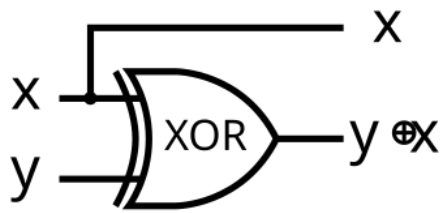
$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Applied to the general superposition

$$CNOT(c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} c_0 \\ c_1 \\ c_3 \\ c_2 \end{bmatrix}$$

The CNOT Gate is the Quantum XOR

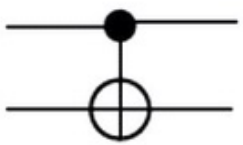
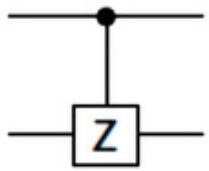




input		output	
x	y	x	y+x
$\|0\rangle$	$\|0\rangle$	$\|0\rangle$	$\|0\rangle$
$\|0\rangle$	$\|1\rangle$	$\|0\rangle$	$\|1\rangle$
$\|1\rangle$	$\|0\rangle$	$\|1\rangle$	$\|1\rangle$
$\|1\rangle$	$\|1\rangle$	$\|1\rangle$	$\|0\rangle$

input		output	
x	y	x	y+x
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

Two Qubit Gates

GATE	CIRCUIT REPRESENTATION	MATRIX REPRESENTATION	TRUTH TABLE
Controlled-NOT gate: apply an X-gate to the target qubit if the control qubit is in state $\|1\rangle$		$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$	Input / Output $\|00\rangle$ / $\|00\rangle$ $\|01\rangle$ / $\|01\rangle$ $\|10\rangle$ / $\|11\rangle$ $\|11\rangle$ / $\|10\rangle$
Controlled-phase gate: apply a Z-gate to the target qubit if the control qubit is in state $\|1\rangle$		$\text{CPHASE} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$	Input / Output $\|00\rangle$ / $\|00\rangle$ $\|01\rangle$ / $\|01\rangle$ $\|10\rangle$ / $\|10\rangle$ $\|11\rangle$ / $-\|11\rangle$

Quantum Teleportation

Teleportation Setup

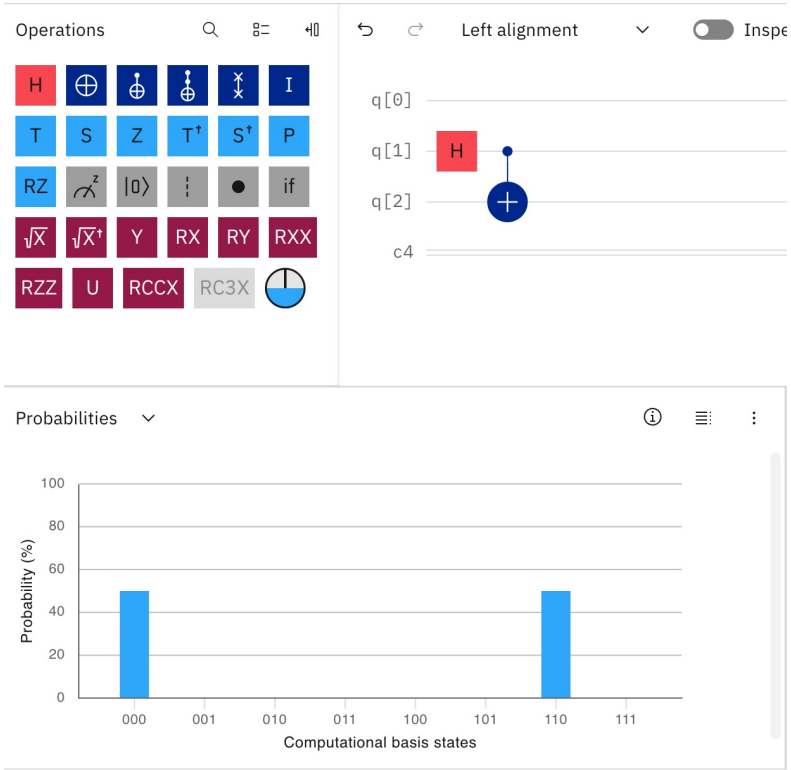
Alice has a qubit (q_0) whose state she wants to be able to send to Bob. Bob has two entangled qubits (q_1 and q_2), he gives q_1 to Alice.

The entangled qubits are in a Bell State of β_{00}:

$$\beta_{00} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

The entire system is therefore in a state:

$$|\psi_0\rangle = |\psi\rangle\,|\beta_{00}\rangle$$





Initial State

Expanding out the initial state:

$$
\begin{aligned}
|\psi_0\rangle &= |\psi\rangle\,|\beta_{00}\rangle \\
&= (\alpha\,|0\rangle + \beta\,|1\rangle)(\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)) \\
&= \frac{1}{\sqrt{2}}\left[(\alpha\,|0\rangle\,(|00\rangle + |11\rangle) + (\beta\,|1\rangle\,(|00\rangle + |11\rangle)\right]
\end{aligned}
$$

By convention, we consider the first two qubits belonging to alice

CNOT + Hadamard Math



We next apply a CNOT gate to q_1 controlled by q_0:

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}\left[(\alpha\,|0\rangle\,(|00\rangle + |11\rangle) + (\beta\,|1\rangle\,(|10\rangle + |01\rangle)\right]$$

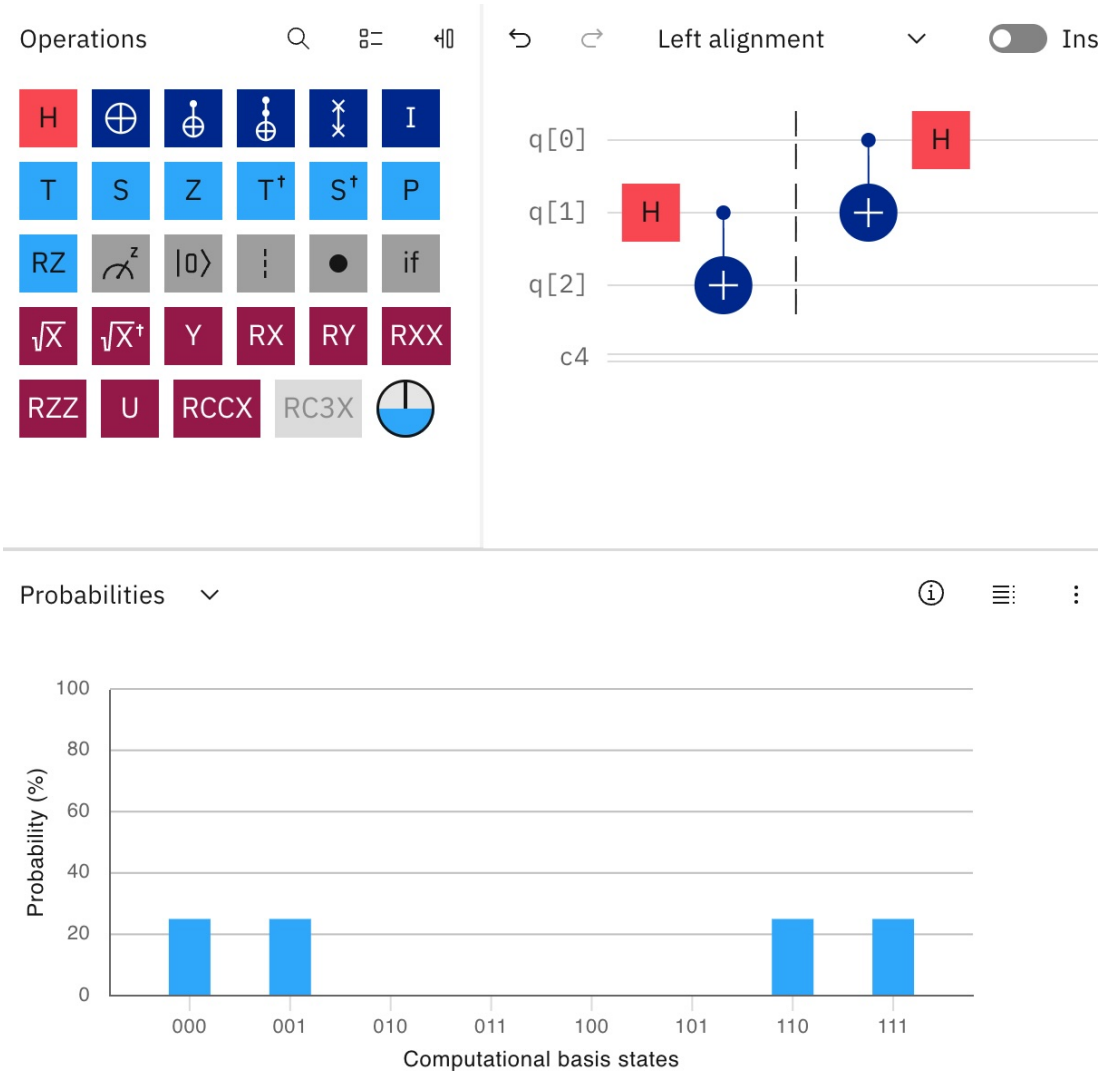
Followed by a Hadamard Gate

- $H\,|0\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$
- $H\,|1\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$

This results in:

$$|\psi_2\rangle = \frac{1}{2}[\alpha(|0\rangle + |1\rangle)(|00\rangle + |11\rangle) + \beta(|0\rangle - |1\rangle)(|10\rangle + |01\rangle)]$$

CNOT + Hadamard Graphically





By Regrouping terms...

Using the associative property:

$$|\psi_3\rangle = \frac{1}{2}[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)]$$

The first belong to Alice and the one in parenthesis belong to Bob. If Alice measures her qubits, there are four possibilities:

$$00 \rightarrow |\psi_3(00)\rangle \equiv \alpha|0\rangle + \beta|1\rangle$$

$$01 \rightarrow |\psi_3(01)\rangle \equiv \alpha|1\rangle + \beta|0\rangle$$

$$10 \rightarrow |\psi_3(10)\rangle \equiv \alpha|0\rangle - \beta|1\rangle$$

$$11 \rightarrow |\psi_3(11)\rangle \equiv \alpha|1\rangle - \beta|0\rangle$$

Alice Measures, Bob then ...

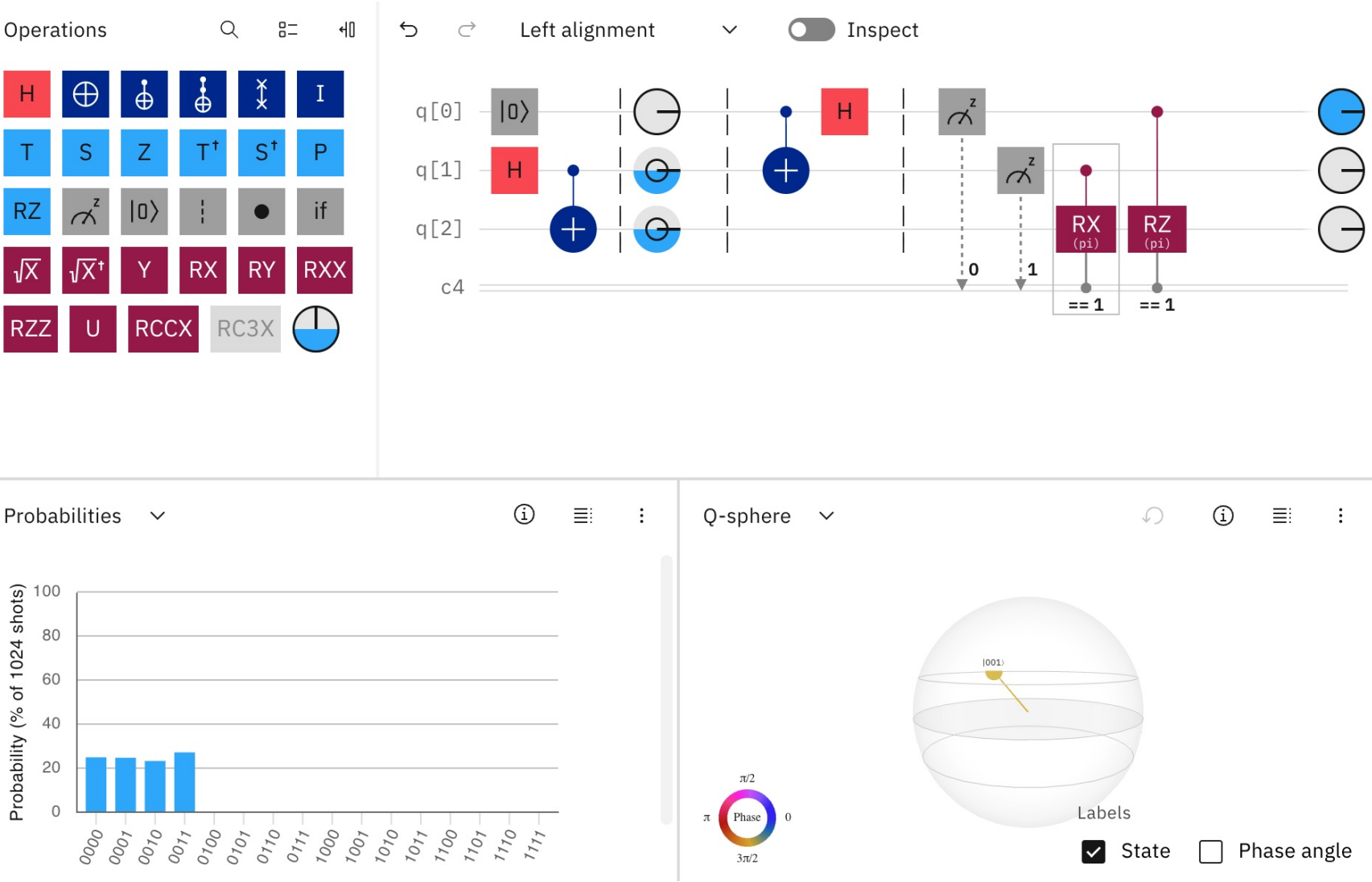
We want Bob's qubit to match $\psi = \alpha|0\rangle + \beta|1\rangle$. Alice measures here two qubits and classically sends her two results to Bob classically. Bob them applies the following gates:

- $00 \rightarrow |\psi_3(00)\rangle \equiv \alpha|0\rangle + \beta|1\rangle \rightarrow$ Bob does nothing
- $01 \rightarrow |\psi_3(01)\rangle \equiv \alpha|1\rangle + \beta|0\rangle \rightarrow$ Bob applies Pauli-X gate
- $10 \rightarrow |\psi_3(10)\rangle \equiv \alpha|0\rangle - \beta|1\rangle \rightarrow$ Bob applies Pauli-Z gate
- $11 \rightarrow |\psi_3(11)\rangle \equiv \alpha|1\rangle - \beta|0\rangle \rightarrow$ Bob applies Pauli-X and Pauli-Z.

Bob now has a qubit that matches Alice's original $|\psi_0\rangle$ and he can be extremely far away from Alice when she does her measurements. This leads to the following two questions:

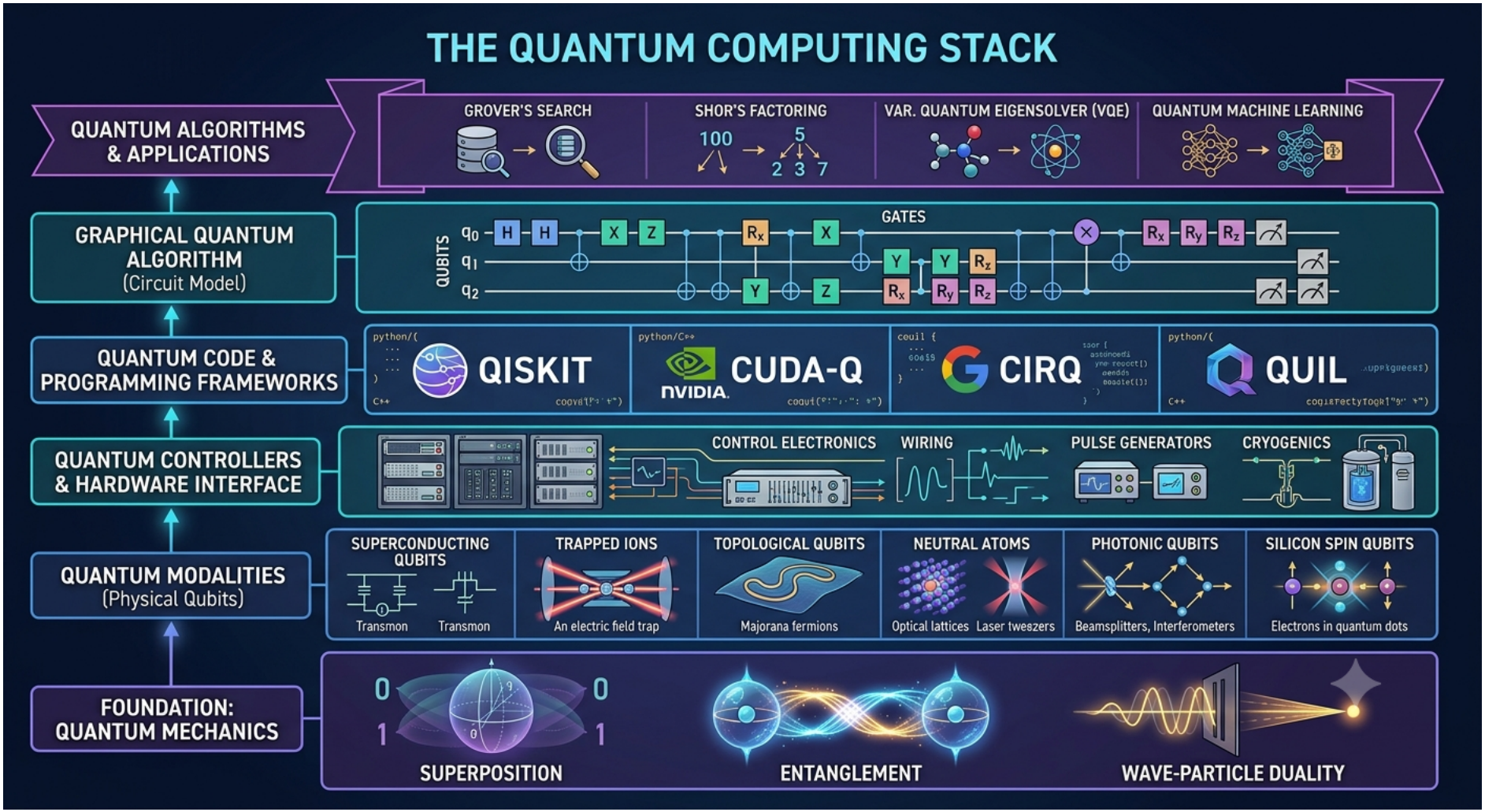
- Did this violate the no cloning theorem?
- Did this violate relativity, information can't be transmitted faster than the speed of light?

Quantum Teleportation



Quantum Computing

Quantum Stack





Types of Quantum Computers

- Superconducting
- Photonic
- Neutral Atom
- Trapped Ion
- Quantum Dots
- Diamond Nitrogen Vacancies

Natural vs Manmade

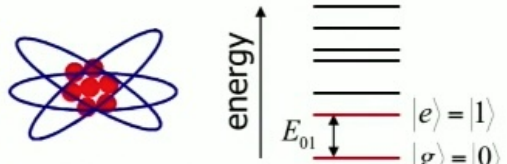
Nature's quantum bits

true qubits



single spin-1/2

effective qubits



single atom

Manmade quantum bits

artificial atoms built from circuits

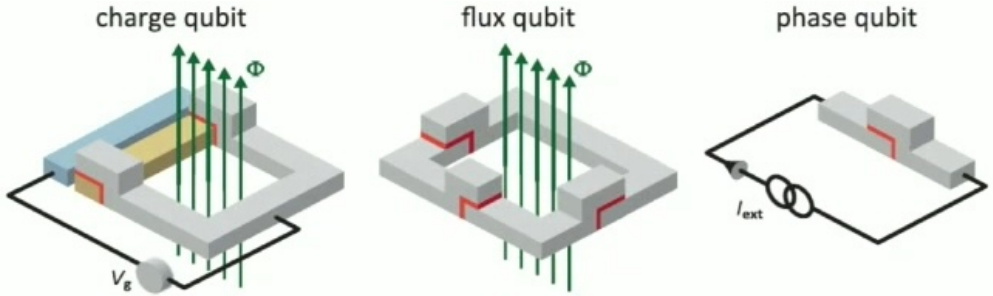


Image credit: W.D. Oliver & P.B. Welander, MRS Bulletin **38**, 816 (2013)

Quantum Computing: Superconducting

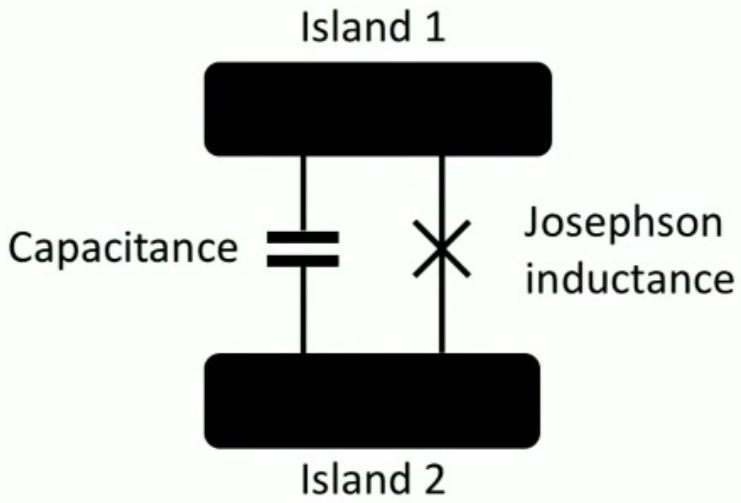


One of the most popular types of quantum computers is a superconducting qubit quantum computer. Usually made from superconducting materials, these quantum computers utilize tiny electrical circuits to produce and manipulate qubits. When using superconducting qubits, gate operations can be performed quickly.
Companies actively researching and manufacturing superconducting quantum computers include Google, IBM, IQM and Rigetti Computing to name just a few.



Superconducting Explored: Transmon QuBit

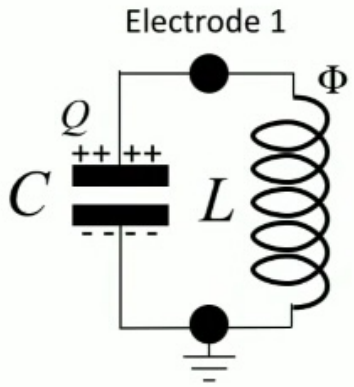
The Transmon qubit



Theory of the transmon: J. Koch *et al.*, Phys Rev. A **76**, 042319 (2007)

Superconducting Explored: LC Oscillator

The quantized *LC* oscillator



Hamiltonian:

$$\hat{H}_{\mathrm{LC}} = \frac{\hat{Q}^2}{2C} + \frac{\hat{\Phi}^2}{2L}$$

Capacitive term — Inductive term

Canonically conjugate variables:

$\hat{\Phi}$ = Flux through the inductor.

$\hat{Q}$ = Charge on capacitor plate.

$$\left[\hat{\Phi}, \hat{Q}\right] = i\hbar$$

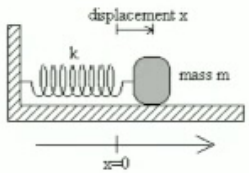
M. Devoret, Les Houches Session LXIII (1995)

Superconducting Explored: Harmonic Oscillator

Correspondence with simple harmonic oscillator

$$\hat{H}_{\mathrm{LC}} = \frac{\hat{\Phi}^2}{2L} + \frac{\hat{Q}^2}{2C} \qquad \left[\hat{\Phi}, \hat{Q}\right] = i\hbar$$

$$\hat{H}_{\mathrm{SHO}} = \frac{k\hat{X}^2}{2} + \frac{\hat{P}^2}{2m} \qquad \left[\hat{X}, \hat{P}\right] = i\hbar$$



Correspondence:

$$\hat{\Phi} \leftrightarrow \hat{X} \qquad L \leftrightarrow \frac{1}{k}$$

$$\hat{Q} \leftrightarrow \hat{P} \qquad C \leftrightarrow m$$

$$\omega = \frac{1}{\sqrt{LC}} \leftrightarrow \sqrt{\frac{k}{m}}$$

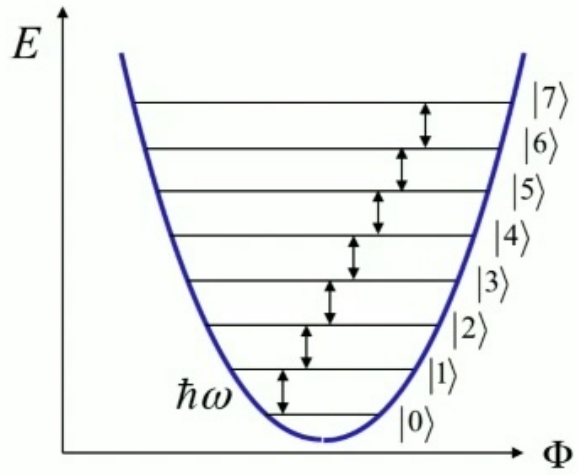
Solve using ladder operators:

$$\hat{a} = \left(\frac{\hat{Q}}{Q_{\mathrm{zpf}}} - i\frac{\hat{\Phi}}{\Phi_{\mathrm{zpf}}}\right) \qquad \Phi_{\mathrm{zpf}} = \sqrt{2\hbar Z}$$

$$\hat{a}^\dagger = \left(\frac{\hat{Q}}{Q_{\mathrm{zpf}}} + i\frac{\hat{\Phi}}{\Phi_{\mathrm{zpf}}}\right) \qquad Q_{\mathrm{zpf}} = \sqrt{2\hbar / Z}$$

$$Z = \omega L = \frac{1}{\omega C} = \sqrt{\frac{L}{C}}$$

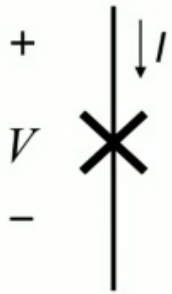
$$\hat{H}_{\mathrm{LC}} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right) \qquad \left[\hat{a}_{\mathrm{r}}, \hat{a}_{\mathrm{r}}^{\dagger}\right] = 1$$



M. Devoret, Les Houches Session LXIII (1995)

Superconducting Explored: Josephson Junction

The Josephson junction



$$I = I_c \sin\left(2\pi \frac{\Phi}{\Phi_o}\right)$$

$$V = \Phi\dot{}$$

$$\Phi_o = \frac{h}{2e}$$

flux quantum

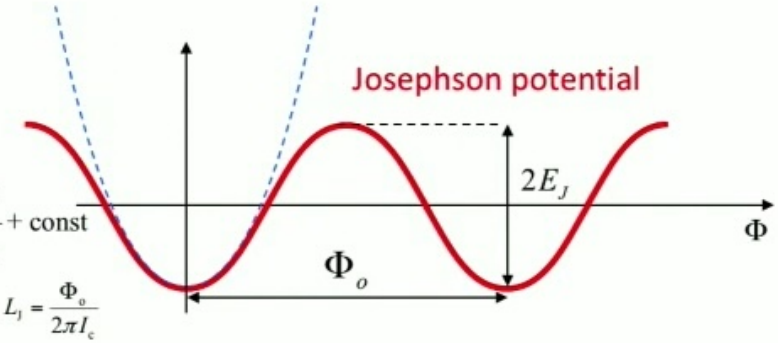


superconductor-insulator-superconductor tunnel junction

$$I_c = \frac{\pi}{2e}\frac{\Delta}{R}$$

$$E_{stored} = E_J\left(1 - \cos\left(2\pi \frac{\Phi}{\Phi_o}\right)\right)$$

$$E_J = \frac{I_c \Phi_o}{2\pi}$$ Josephson Energy



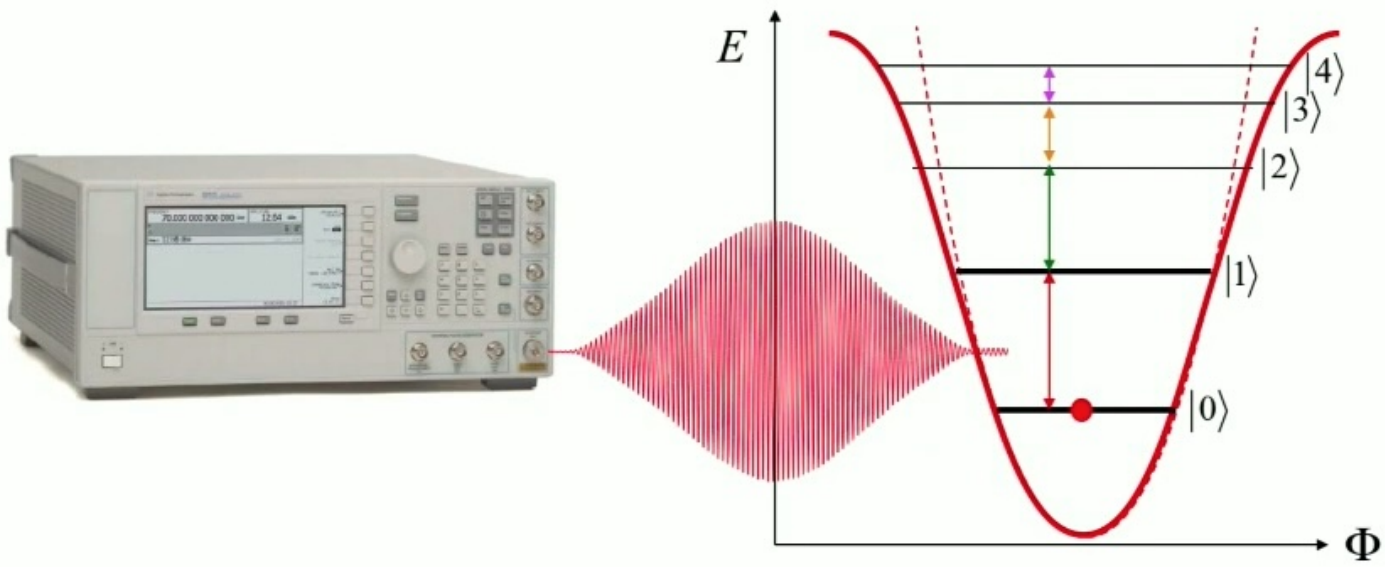
For small flux: $E_J \approx \frac{\Phi^2}{2L_J} + \text{const}$

Josephson inductance $L_J = \frac{\Phi_o}{2\pi I_c}$

M. Devoret, Les Houches Session LXIII (1995)

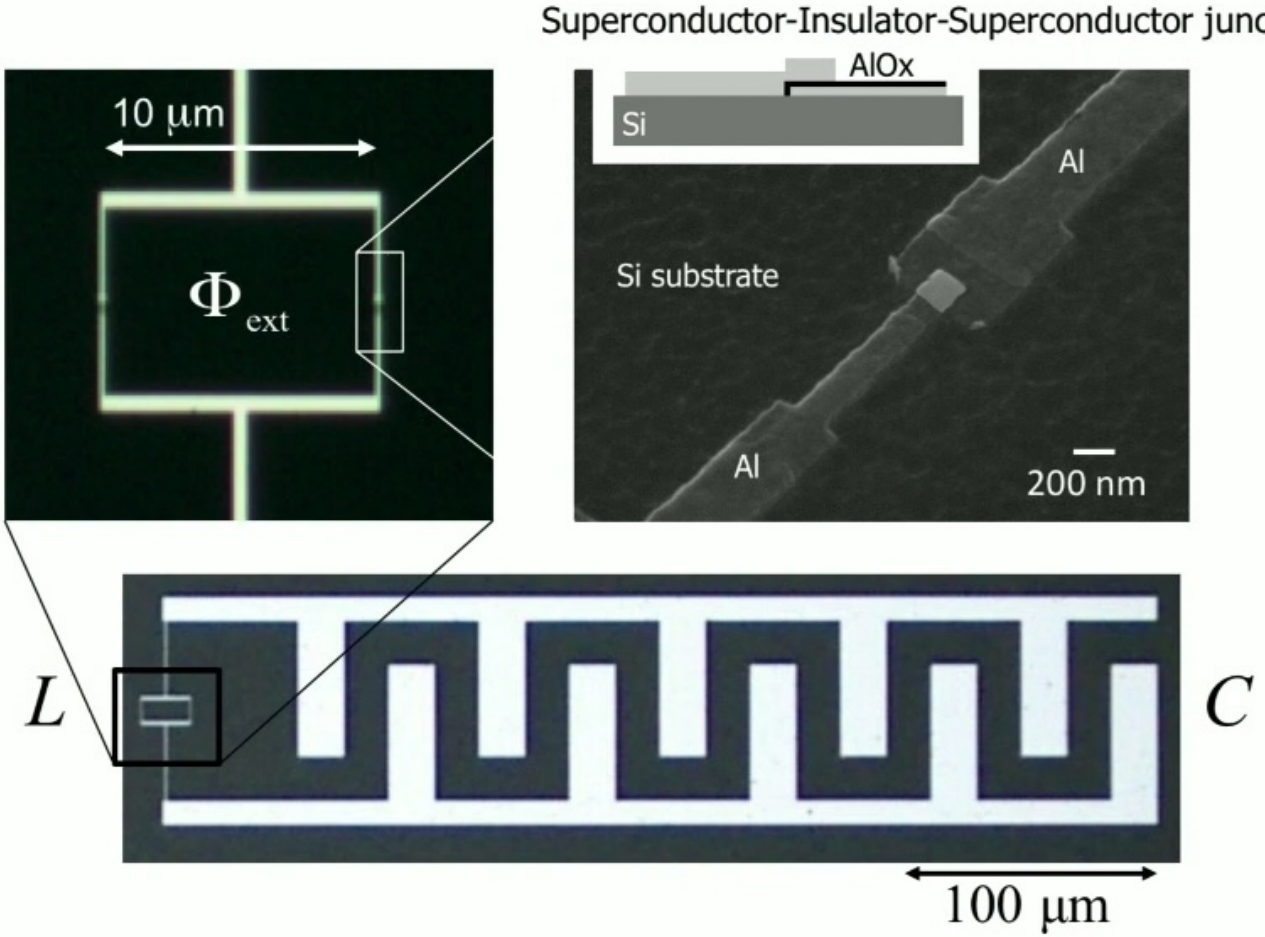
Superconducting Explored: Energy Quantization Spectrum

Transmon energy spectrum



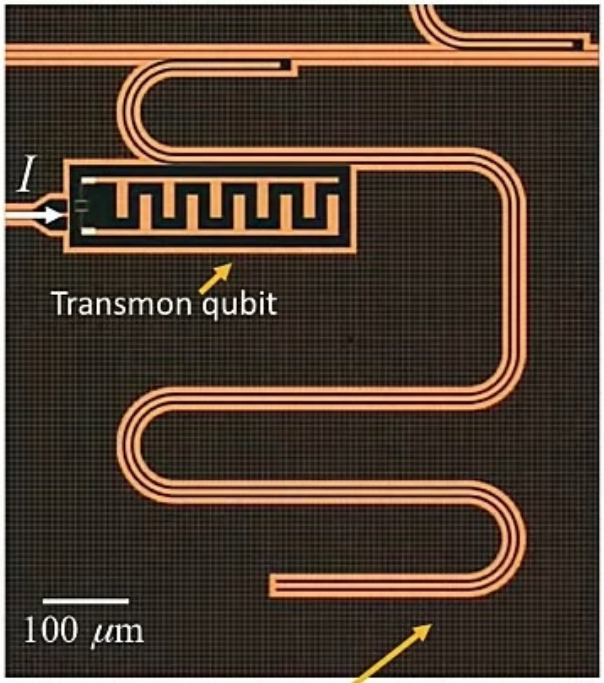
Superconducting Explored: Josephson Junction

Two-junction transmon



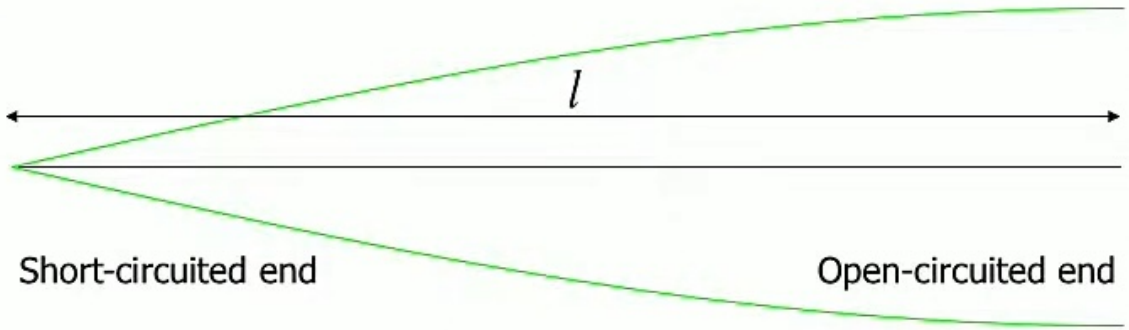
Superconducting Explored: QuBit Resonator

Qubit-resonator interaction



Resonator modes (usually we focus on fundamental) are set by

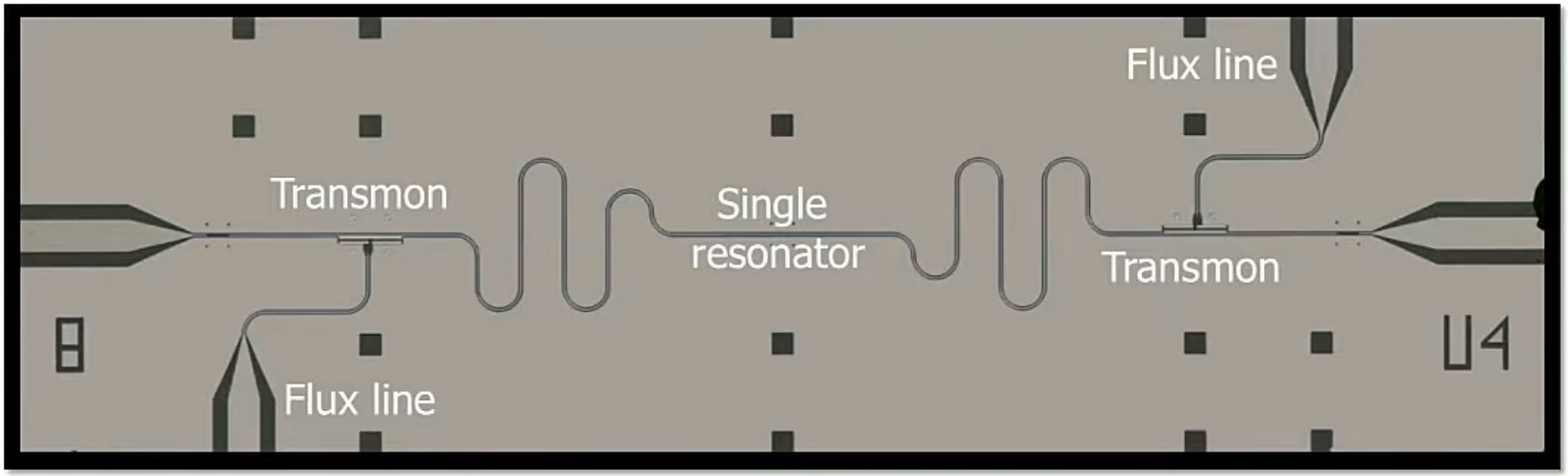
the total resonator length l
the end terminations (short or open circuit)
the phase velocity (slower than speed of light in vacuum by factor 2-3)



The coupling between qubit and resonator is typically capacitive

Superconducting Explored: Two QuBit

First-generation two-qubit quantum processor

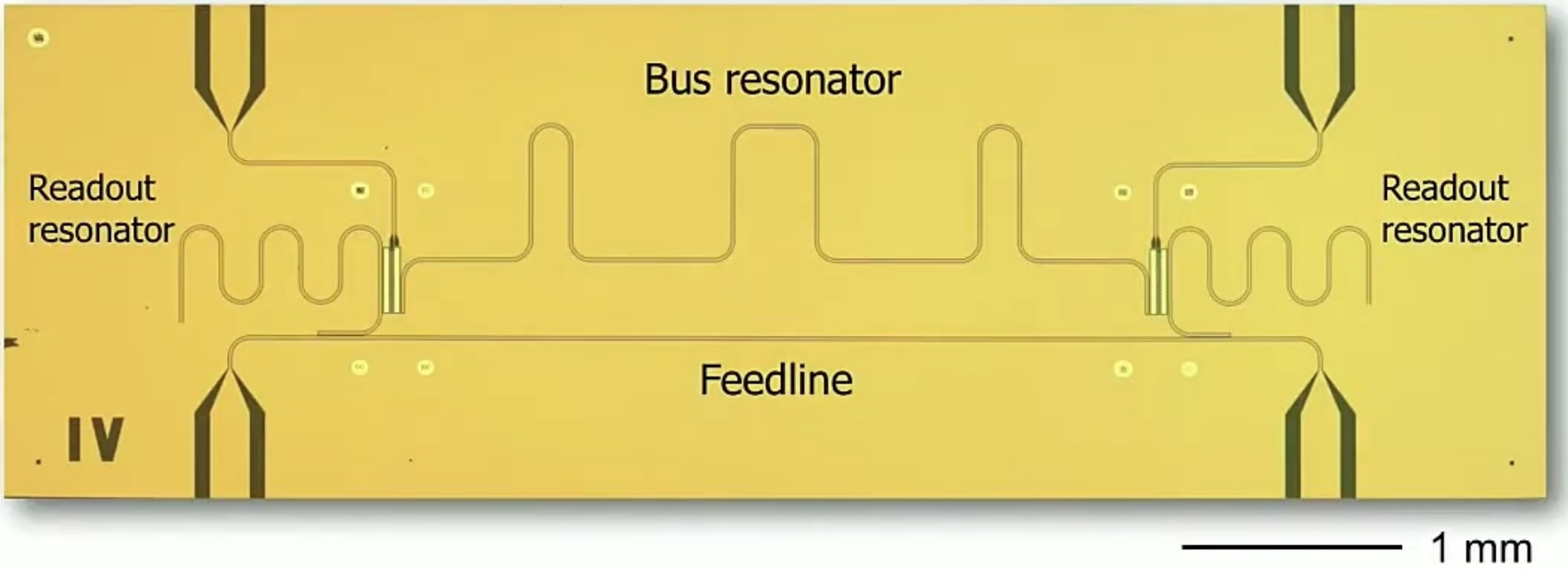


One resonator for both qubit-qubit coupling and for readout

DiCarlo *et al.*, Nature (2009)

Superconducting Explored: Two QuBit Improved

Second-generation two-qubit processor

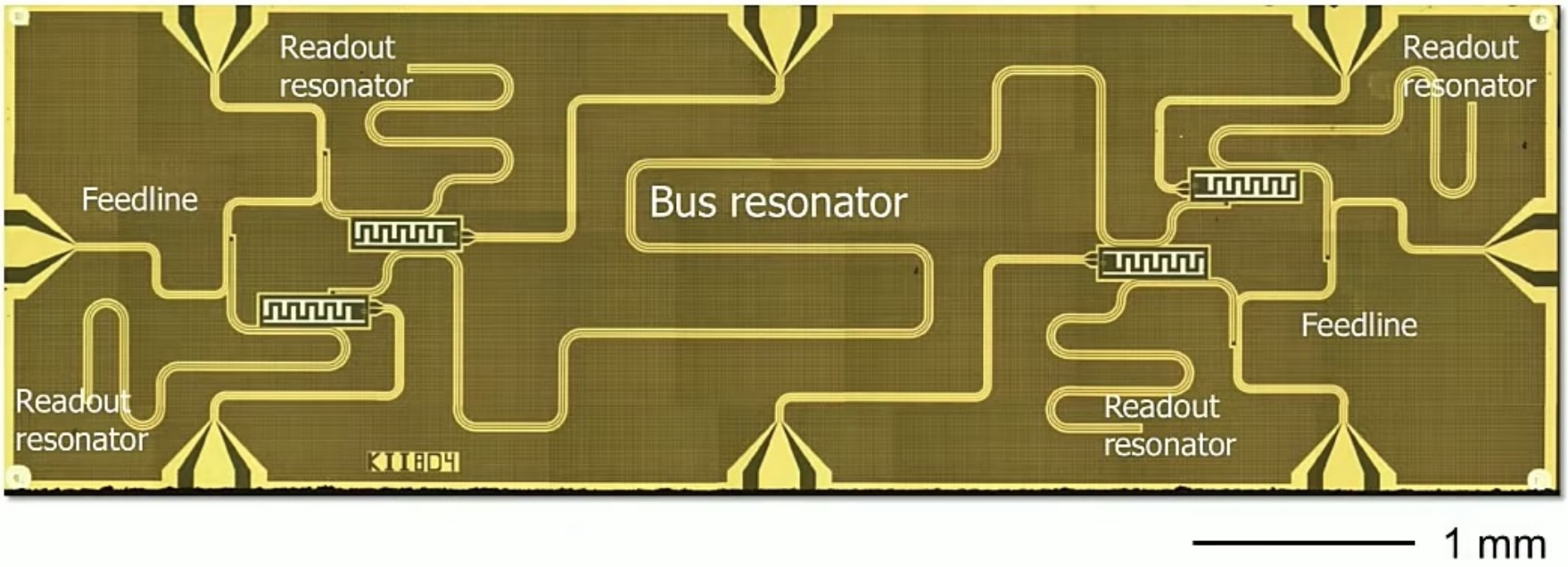


Different resonators for qubit-qubit coupling and for readout of each qubit

Groen *et al.*, Phys. Rev. Lett. (2013)

Superconducting Explored: Four QuBit

Second-generation four-qubit processor



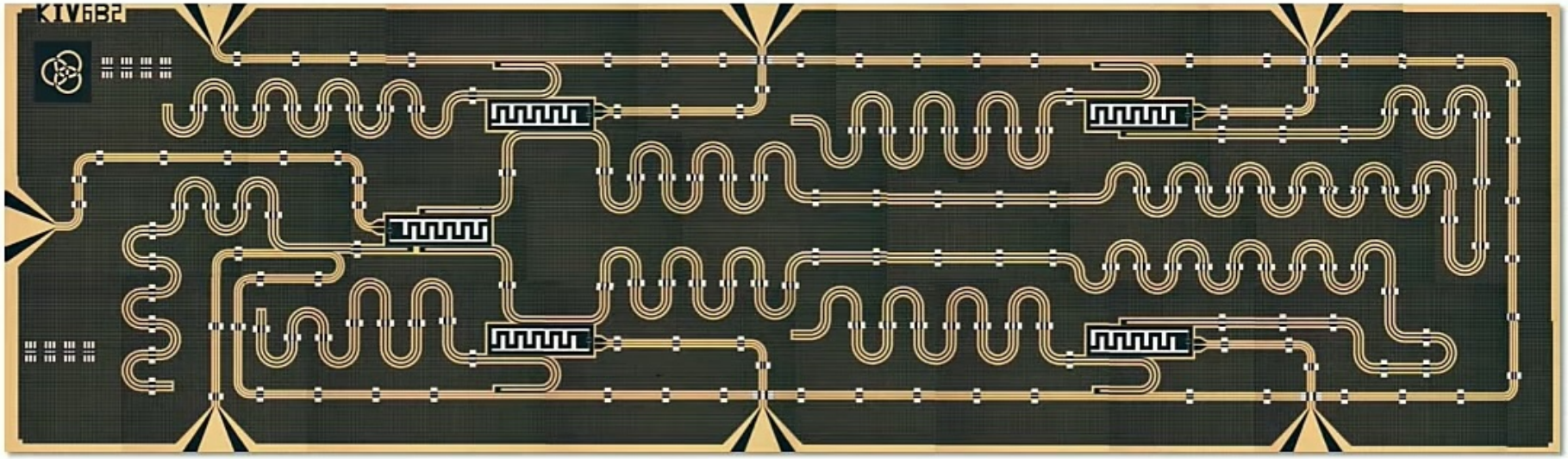
Different resonators for qubit-qubit coupling and for readout of each qubit

Saira *et al.*, Phys. Rev. Lett. (2014)

Superconducting Explored: Four QuBit

Second-generation five-qubit processor

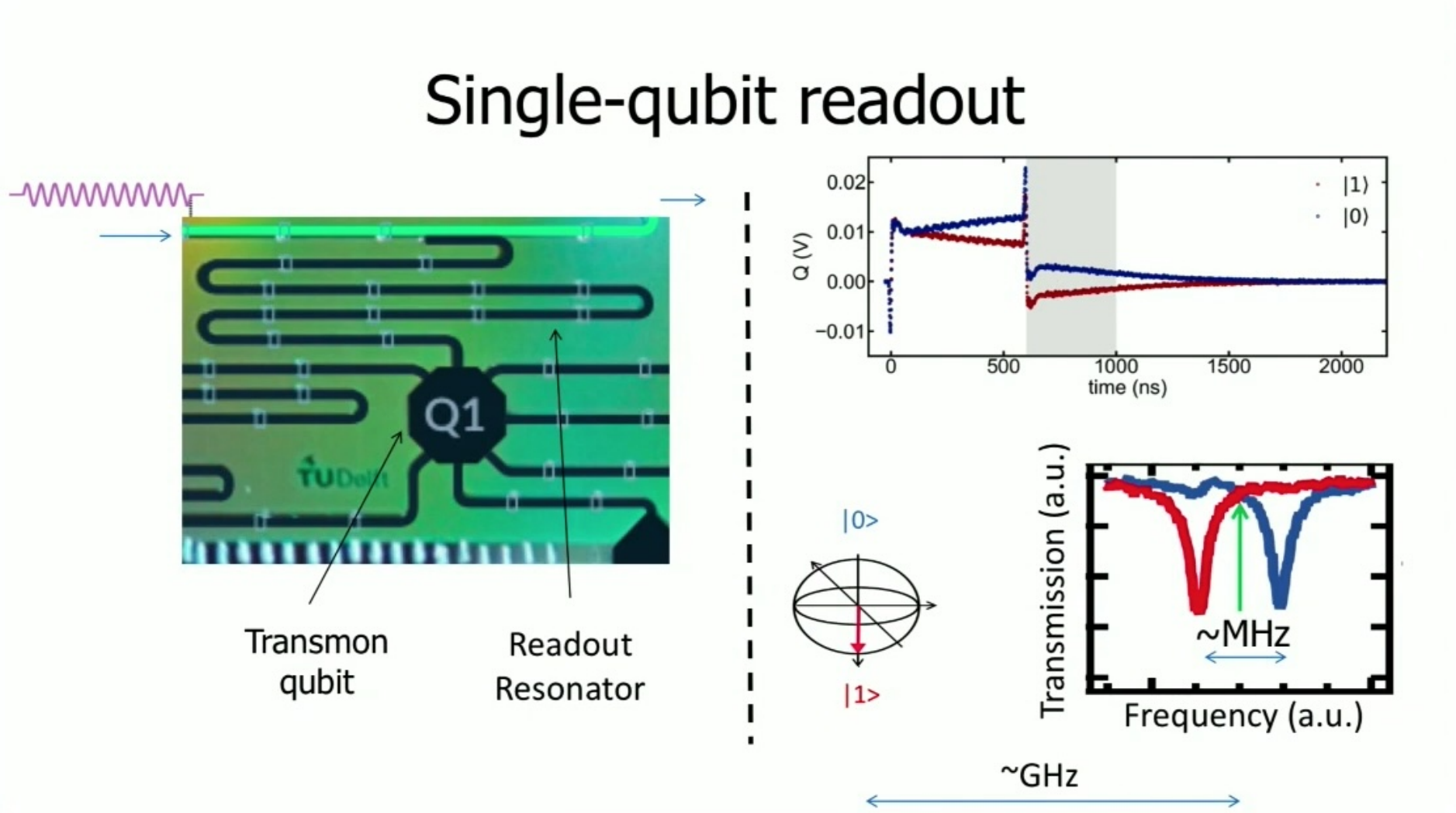
Challenge: identify the readout resonators and bus resonators



Different resonators for qubit-qubit coupling and for readout of each qubit

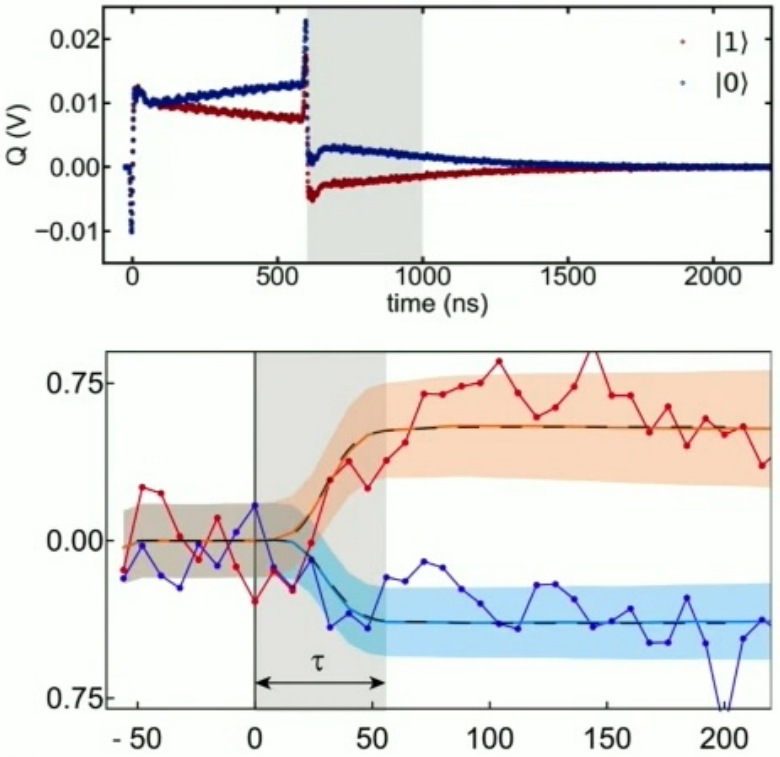
Riste *et al.*, Nature Communications (2015)

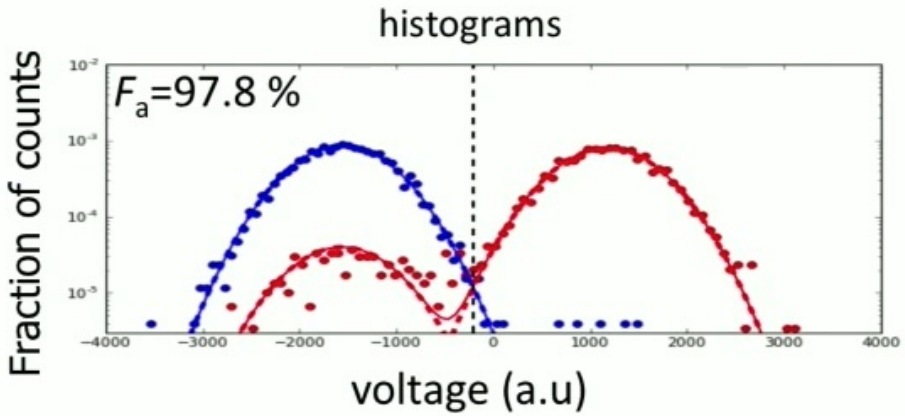
Superconducting Explored: Readout



Superconducting Explored: Readout Noise

Single-qubit readout

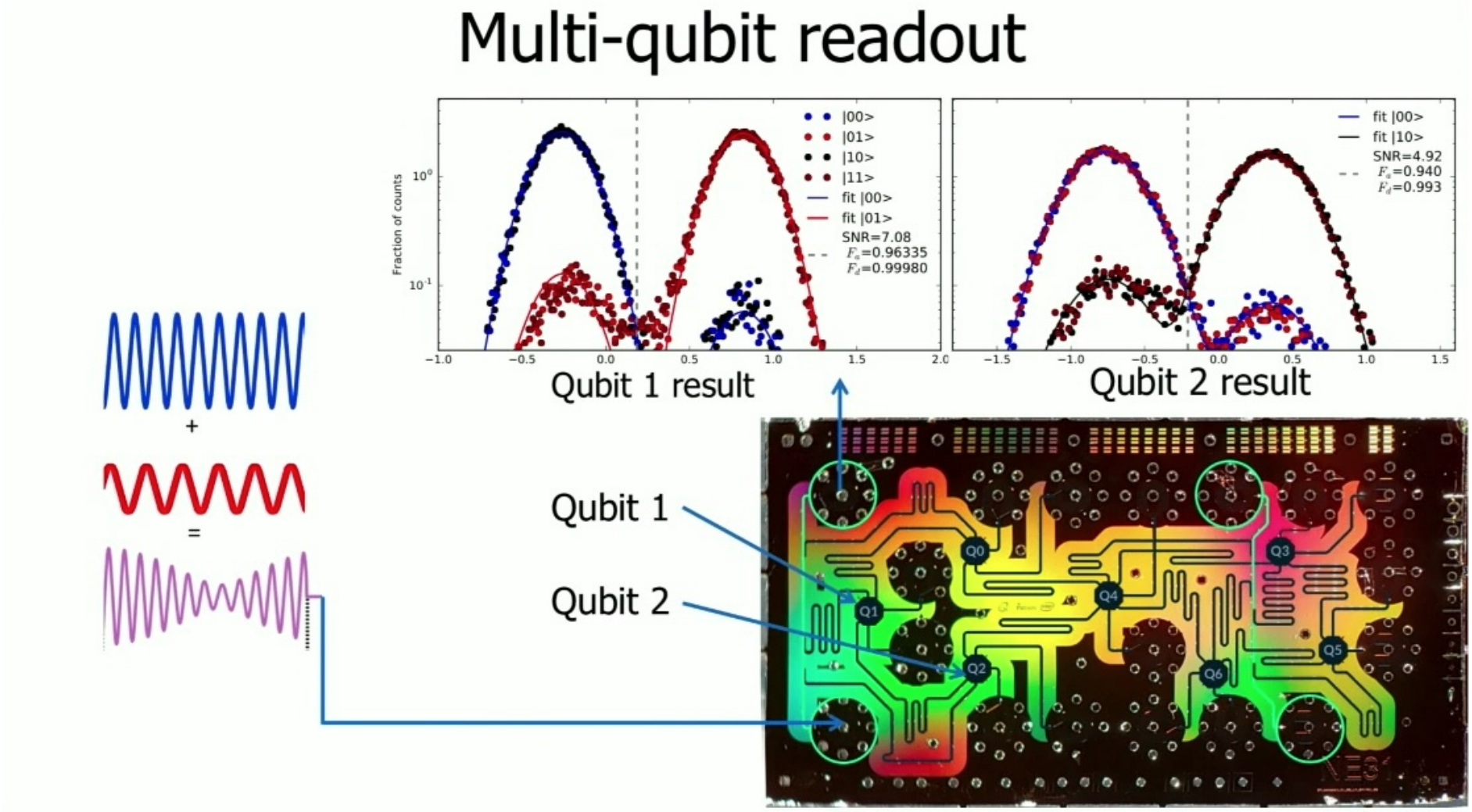




$$\mathcal{F}_{\mathrm{a}} = 1 - \frac{1}{2}\left(\epsilon_{01} + \epsilon_{10}\right)$$

[1]: D. Ristè *et al.*, PRL 109, 050507 (2012)
[2]: T. Walter *et al.*, PR Appl. 7, 054020 (2017)

Superconducting Explored: Two QuBit Readout



Quantum Computing: Photonics



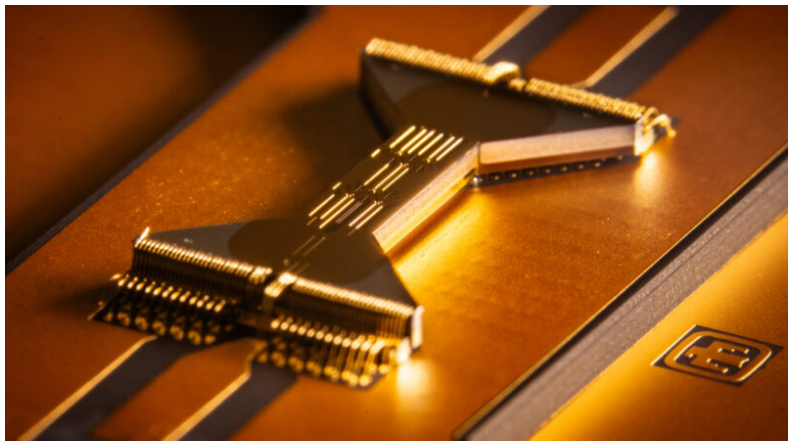
These types of quantum computers use photons (particles of light) to carry and process quantum information. For large-scale quantum computers, photonic qubits are a promising alternative to trapped ions and neutral atoms that require cryogenic or laser cooling.

Quantum Computing: Neutral Atom



Quantum computing based on neutral atoms involves atoms suspended in an ultrahigh vacuum by arrays of tightly focused laser beams called optical tweezers, though not all neutral atom companies use optical tweezers. Neutral atom quantum computers are less sensitive to stray electric fields, which makes them a good option for quantum processors.

Trapped Ions



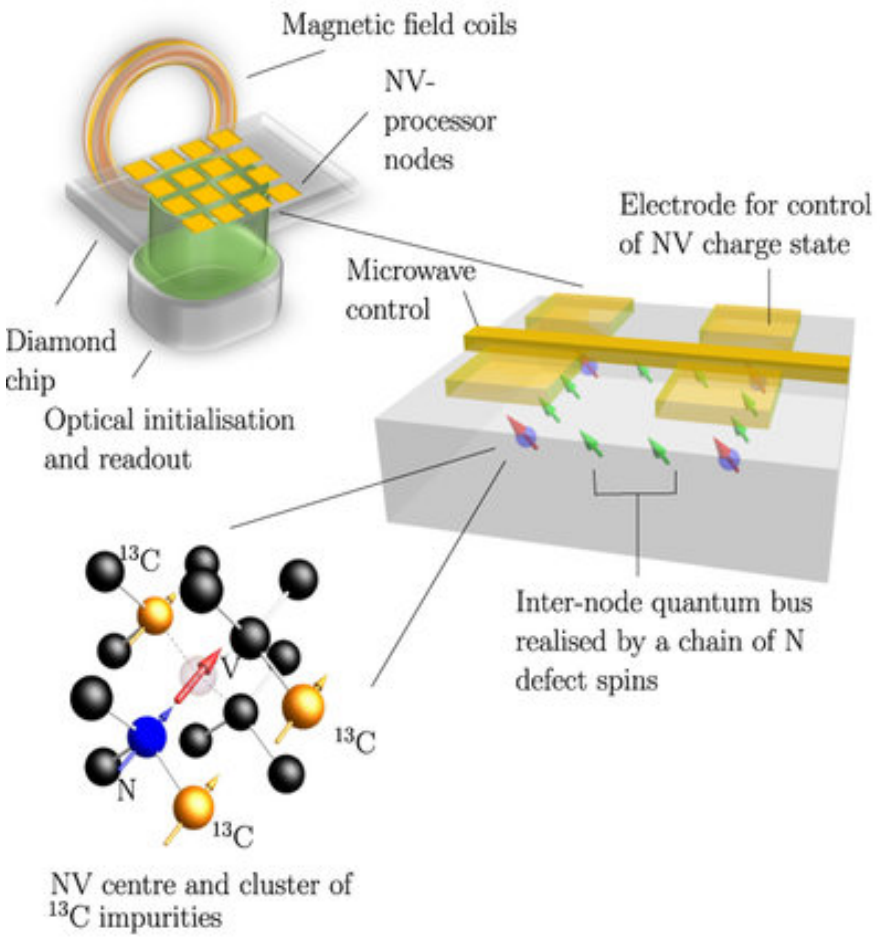
A trapped ion quantum computer involves using atoms or molecules with a net electrical charge known as “ions” that are trapped and manipulated using electric and magnetic fields to store and process quantum information. As trapped ions can be isolated from their environment, they are useful for precision measurements and other applications requiring high levels of stability and control. Also, the qubits can remain in a superposition state for a long time before becoming decoherent. Representing the trapped ions community of companies in the quantum space, we have Quantinuum (a company that came out of the merger between Cambridge Quantum Computing and Honeywell Quantum



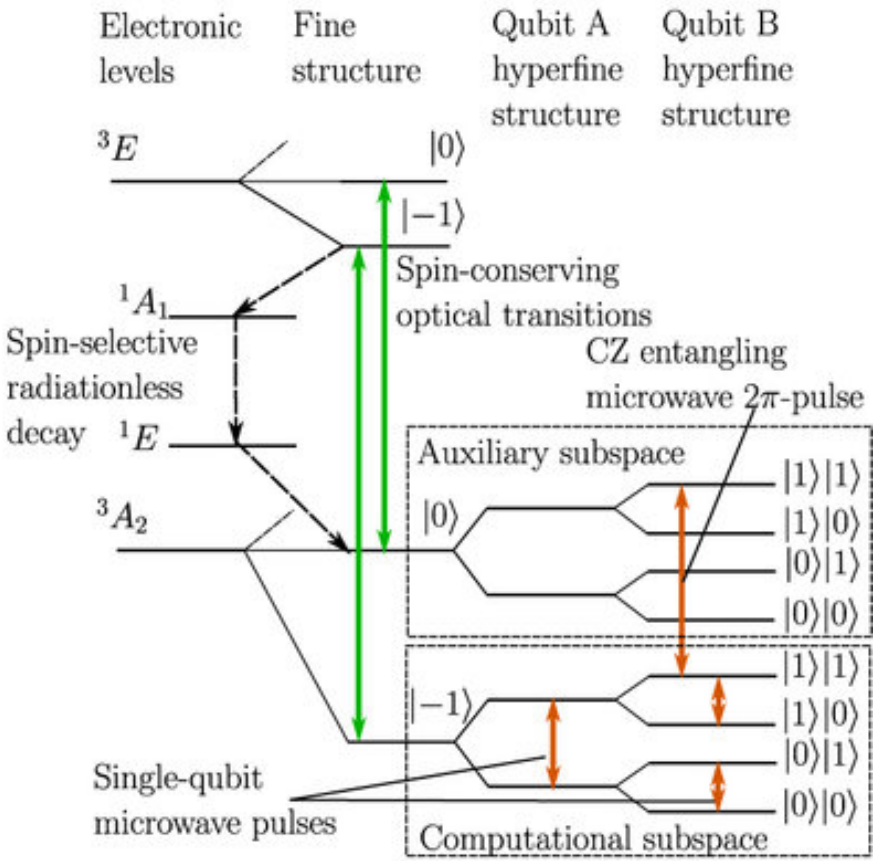
Quantum Computing: Quantum Dots

A quantum dot quantum computer uses silicon qubits made up of pairs of quantum dots. In theory for quantum computers, such 'coupled' quantum dots could be used as robust quantum bits, or qubits.
Companies focused on this area include Diraq, Siquance and Quantum Motion.

Quantum Computing: NV Diamond



(a)



(b)